5 Continuous-Time Fourier Transform

Objective

After completing this chapter, the user will be able to understand

1. The types of Fourier transform
2. Properties of Fourier transform
3. How to find FT of various signals
5. Parseval's theorem
6. Signal transmission through LTI system
8. Eigenvalue and eigenfunction

Introduction

1. In this chapter, we consider Fourier transform for describing both periodic and non-periodic signals for which no Fourier series exists.
2. Like the Fourier series coefficients, the Fourier transform specifies the spectral content of a signal, thus providing a frequency domain description of the signal useful in analytically representing aperiodic signals, the Fourier transform is a valuable tool in the analysis of LTI systems.
3. An aperiodic signal can be viewed as a periodic signal with an infinite period.
4. As the period becomes infinite, the frequency components from the continuum and the Fourier series sum become integral.
5. The resulting spectrum coefficient in this representation is called the Fourier transform.

Continuous-Time Fourier transform:

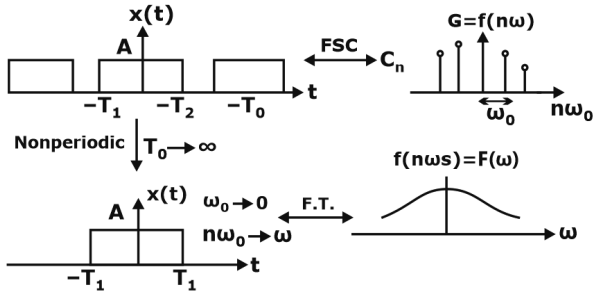


Fig. 5.1

First fig. replace T2 with T1 if it is symmetric.

$$C_n = \frac{1}{T_0}\int_{-T_0/2}^{T_0/2} x(t)e^{-jn\omega_0 t}dt \longrightarrow \frac{X(n\omega_0)}{T_0}$$

$$X(n\omega_0) = \int_{-T_0/2}^{T_0/2} x(t)e^{-jn\omega_0 t}dt$$

$T_0 \to \infty$

$\omega_0 \to 0$

$n\omega_0 \to \omega$

$$X(\omega_0) = \int_{-\infty}^{\infty} x(t)e^{-j\omega t}dt$$

Relations

$x(t) \xrightarrow{F.T.} X(\omega)$

$x(t) \xrightarrow{F.T.} X(j\omega)$

$x(t) \xrightarrow{F.T.} X(f)$

$x(t) \xrightarrow{F.T.} X(jf)$

$x(t) \xrightarrow{F.T.} X(\omega)$

and

$x(t) \xrightarrow{F.T.} X(f)$ *used.*

Conversion between $X(\omega)$ and $X(f)$

Mathematical conversion:

1. $X(\omega) = \omega^2$

   $\downarrow$

   $X(f) = (2\pi f)^2 = 4\pi^2 f^2$

2. $X(\omega) = \delta(\omega)$

   $\downarrow$

   $X(f) = \delta(2\pi f) = \frac{1}{2\pi}\delta(f)$

3. $X(\omega) = \sin\omega$

   $\downarrow$

   $X(f) = \sin 2\pi f$

Graphical conversion:

S.1 $X(\omega) \longrightarrow X(f)$

S.2 Horizontal axis → Divide by 2π

S.3 Vertical axis → Impulse present → Divide by 2π

→ Impulse absent → Keep it as it is

Note: $X(\omega)\big|_{\omega=0} = X(f)\big|_{f=0}$

Example 5.1:

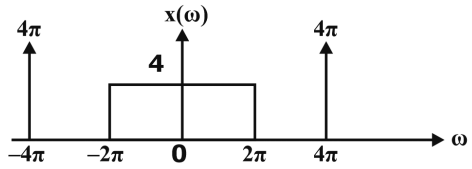


Fig. 5.2

Solution:

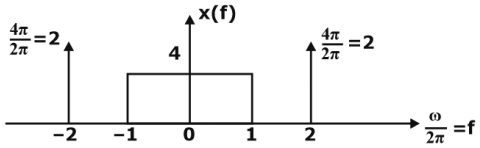


Fig. 5.3

Types of FT Well defined FT (WDFT):

1. Calculated for absolutely integrable signal.
2. Formula of FT can be applied.

$$X(\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt$$

$$X(f) = \int_{-\infty}^{\infty} x(t) e^{-j2\pi ft} dt$$

3. Properties of FT can be applied.
4. WDFT never goes to infinity even for a single value of "ω".

Limitedly defined FT (LDFT):

1. Calculated for power signals.
2. Formula of FT cannot be used.
3. To calculate FTIndirect technique
   a) WDFT results are limited
4. Properties can be applied.
5. LDFT goes to ∞ for at least one value of ω.

Convergence of Fourier Transform

Dirichlet showed that if x(t) satisfies certain conditions (Dirichlet conditions), its Fourier transform is guaranteed to converge pointwise at all points where x(t) is continuous. These conditions are as follows:

1. x(t) be absolutely integrable, i.e.,

$$\int_{-\infty}^{\infty} |x(t)| dt < \infty$$

This guarantees that X(ω) will be finite since

$$|X(\omega)| = \left| \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt \right|$$

$$|X(\omega)| \le \int_{-\infty}^{\infty} |x(t) e^{-j\omega t}| dt = \int_{-\infty}^{\infty} |x(t)| dt$$

So

$$\text{if } \int_{-\infty}^{\infty} |x(t)| dt < \infty, \text{ then } |X(\omega)| < \infty$$

2. In any finite interval of time, x(t) is of bounded variation, i.e., x(t) has a finite number of maxima and minima.
3. In any finite interval of time, there are only a finite number of discontinuities. Furthermore, each of these discontinuities is finite.

$$X(\omega_0) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt$$

FT converges or FT exist or FT is finite:

$$X(\omega): converge$$

$$\downarrow$$

$$X(\omega) = |X(\omega)| e^{j\angle x(\omega)}$$

$$\downarrow$$

$$|X(\omega)|: converge$$

$$|X(\omega_0)| = \left| \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt \right| \le \int_{-\infty}^{\infty} |x(t)| |e^{-j\omega t}| dt$$

$$|X(\omega_0)| \le \int_{-\infty}^{\infty} |x(t)| dt$$

$$\downarrow \qquad \downarrow$$

$$finite \qquad finite$$

$$\downarrow$$

$$F.T. converges$$

Formula:

$$x(t) \xleftrightarrow{F.T.} X(\omega)$$

$$X(\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt$$

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega) e^{j\omega t} d\omega$$

$$x(t) \longleftrightarrow X(f)$$

$$X(f) = \int_{-\infty}^{\infty} x(t) e^{-j2\pi ft} dt$$

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(f) e^{j2\pi ft} df$$

$$\omega = 2\pi f$$

$$d\omega = 2\pi df$$

Graph:

$$\begin{cases} \rightarrow X(\omega) \text{ vs } \omega \\ \rightarrow X(\omega) = |X(\omega)| e^{j\angle X(\omega)} \end{cases}$$

$$\begin{cases} \rightarrow X(f) \text{ vs } f \\ \rightarrow X(f) = |X(f)| e^{j\angle X(f)} \end{cases}$$

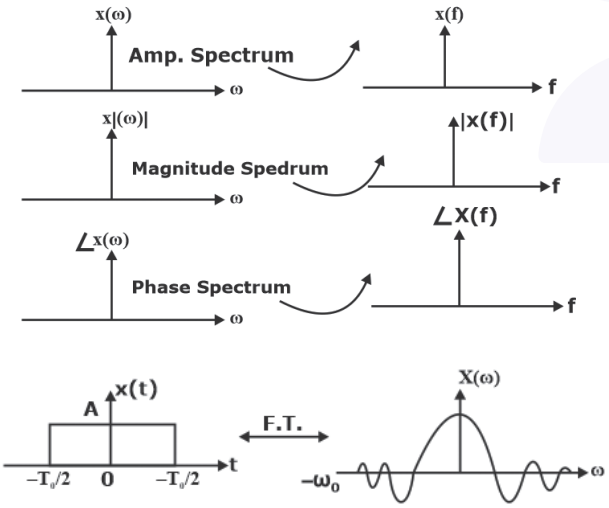


Fig. 5.4

Properties of FT:Linearity:

$$x_1(t) \longleftrightarrow X_1(\omega) \quad x_1(t) \longleftrightarrow X_1(f)$$

$$x_2(t) \longleftrightarrow X_2(\omega) \quad x_2(t) \longleftrightarrow X_2(f)$$

$$Ax_1(t) + Bx_2(t) \longleftrightarrow AX_1(\omega) + BX_2(\omega)$$

$$Ax_1(t) + Bx_2(t) \longleftrightarrow AX_1(f) + BX_2(f)$$

Time shifting:

$$x(t) \longleftrightarrow X(\omega) \quad x(t) \longleftrightarrow X(f)$$

$$x(t - t_0) \longleftrightarrow e^{-j\omega t_0} X(\omega)$$

$$x(t - t_0) \longleftrightarrow e^{-j2\pi f t_0} X(f)$$

$$x(t + t_0) \longleftrightarrow e^{j\omega t_0} X(\omega)$$

$$x(t + t_0) \longleftrightarrow e^{j2\pi f t_0} X(f)$$

Extensions:

$$(1) \quad \frac{x(t + t_0) + x(t - t_0)}{2} \longleftrightarrow \cos(t_0)\omega X(\omega) = \cos(t_0\omega) X(\omega)$$

$$(2) \quad \frac{x(t + t_0) - x(t - t_0)}{2} \longleftrightarrow \sin(t_0)\omega X(\omega) = \sin(t_0\omega) X(\omega)$$

Example 5.2:

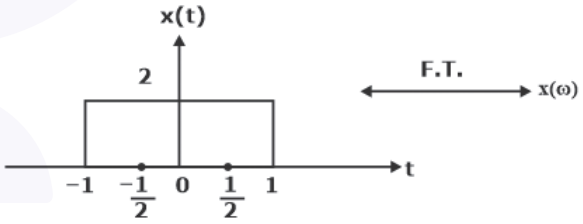


Fig. 5.5

$$G(\omega) = X(\omega) \cos\frac{\omega}{2}$$

$$g(t) \longleftrightarrow G(\omega)$$

$$g(t)\big|_{t=0} = ?$$

Solution: $G(\omega) = X(\omega) \cos\frac{\omega}{2}$

$$G(\omega) = \frac{X(\omega) e^{j\omega/2} + X(\omega) e^{-j\omega/2}}{2} \leftrightarrow g(t)$$

$$= \frac{x\left(t + \frac{1}{2}\right) + x\left(t - \frac{1}{2}\right)}{2}$$

$$x(t) \longleftrightarrow X(\omega)$$

$$x\left(t+\frac{1}{2}\right) \longleftrightarrow e^{j\omega/2}X(\omega)$$

$$x\left(t-\frac{1}{2}\right) \longleftrightarrow e^{-j\omega/2}X(\omega)$$

$$g(t=0)=\frac{x\left(\frac{1}{2}\right)+x\left(-\frac{1}{2}\right)}{2}$$

$$g(0)=2$$

Note: $x(t) \longleftrightarrow X(\omega)=\left|X(\omega)\right|e^{j\angle X(\omega)}$

$$g(t)=x(t\pm t_0) \longleftrightarrow G(\omega)=e^{\pm j\omega t_0}X(\omega)$$

$$\left|G(\omega)\right|e^{j\angle G(\omega)}=G(\omega)=\left|X(\omega)\right|e^{j\left[\angle X(\omega)\pm\omega t_0\right]}$$

$$\left|G(\omega)\right|=\left|X(\omega)\right|$$

$$\angle G(\omega)=\angle X(\omega)\pm\omega t_0$$

Magnitude remains unchanged.

Frequency shifting:

$$x(t) \longleftrightarrow X(\omega) \qquad x(t) \longleftrightarrow X(f)$$

$$e^{j\omega_0 t}x(t) \longleftrightarrow X(\omega-\omega_0)$$

$$e^{j2\pi f_0 t}x(t) \longleftrightarrow X(f-f_0)$$

$$e^{-j\omega_0 t}x(t) \longleftrightarrow X(\omega+\omega_0)$$

$$e^{-j2\pi f_0 t}x(t) \longleftrightarrow X(f+f_0)$$

Extension:

1. $x(t)\cdot\cos\omega_0 t \longleftrightarrow \dfrac{X(\omega-\omega_0)+X(\omega+\omega_0)}{2}$
2. $x(t)\cdot\cos 2\pi f_0 t \longleftrightarrow \dfrac{X(f-f_0)+X(f+f_0)}{2}$
3. $x(t)\cdot\sin\omega_0 t \longleftrightarrow \dfrac{X(\omega-\omega_0)-X(\omega+\omega_0)}{2j}$
4. $x(t)\cdot\sin 2\pi f_0 t \longleftrightarrow \dfrac{X(f-f_0)-X(f+f_0)}{2j}$

Modulation property:

$$\frac{x(t) \longleftrightarrow x(\omega-\omega_0)+x(\omega+\omega_0)}{2}$$

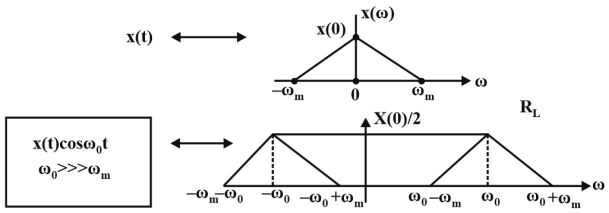


Fig. 5.6

Note: $x(t) \xrightarrow{max.freq} f_m Hz$

$$x(t)\cdot\cos 2\pi f_0 t \xrightarrow{max.freqn} (f_0+f_m)Hz$$

Time reversal:

$$x(t) \longleftrightarrow X(\omega) \qquad x(t) \longleftrightarrow X(f)$$

$$x(-t) \longleftrightarrow X(-\omega) \qquad x(-t) \longleftrightarrow X(-f)$$

Extension:

x(t)	x(ω)/x(f)
Even OddNeno	Even OddNENO

Time scaling:

$$x(t) \longleftrightarrow X(\omega) \qquad x(t) \longleftrightarrow X(f)$$

$$x(at) \longleftrightarrow \frac{1}{|a|}X\left(\frac{\omega}{a}\right) \qquad x(at) \longleftrightarrow \frac{1}{|a|}X\left(\frac{f}{a}\right)$$

$$\begin{cases} x(t) \longleftrightarrow X(\omega) \\ compress \longleftrightarrow Expand \\ Expand \longleftrightarrow compress \end{cases}$$

Example 5.3:

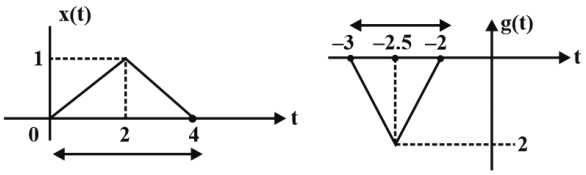


Fig. 5.7

G(f) in terms of X(f) = ?

Solution:

$$g(t) = -2x(4t+12)$$

$$x(t) \longleftrightarrow X(f)$$

$$x(t+12) \longleftrightarrow e^{j2\pi(12)f} X(f)$$

$$x(4t+12) \longleftrightarrow \frac{1}{4} e^{j2\pi(12)f/4} X\left(\frac{f}{4}\right)$$

$$g(t) = -2x(4t+12) \longleftrightarrow \frac{-1}{2} e^{j6\pi f} X\left(\frac{f}{4}\right)$$

Or

$$g(t) = -2x(-4t-8)$$

$$x(t) \longleftrightarrow X(f)$$

$$x(t-8) \longleftrightarrow e^{-j2\pi f(8)} X(f)$$

$$x(4t-8) \longleftrightarrow \frac{1}{4} e^{-j2\pi f(8)/4} X\left(\frac{f}{4}\right)$$

$$g(t) = -2x(-4t+8) \longleftrightarrow$$

$$\left(\frac{1}{4} e^{j4\pi f} X\left(-\frac{f}{4}\right)(-2)\right) = G(f)$$

Fourier Transform of Standard Signals

WDFT:

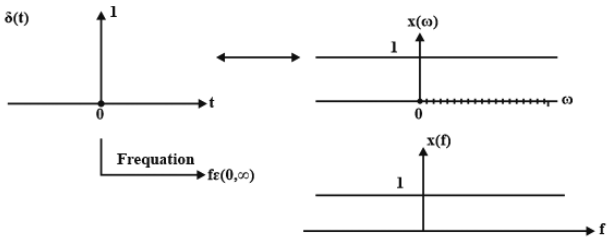
Impulse signal:

$$x(t) = \delta(t)$$

$$X(\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt$$

$$X(\omega) = \int_{-\infty}^{\infty} \delta(t) e^{-j\omega t} dt = 1$$

$$x(t) = \delta(t) \xleftrightarrow{F.T.} X(\omega) = 1$$



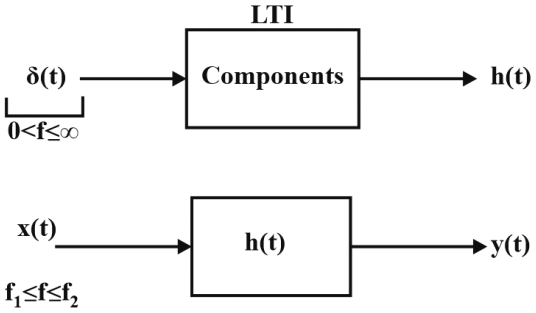


Fig. 5.8

$$\delta(t) \longleftrightarrow 1$$

$$\frac{\delta(t+a)+\delta(t-a)}{2} \longleftrightarrow \cos a\omega$$

$$\frac{\delta(t+a)-\delta(t-a)}{2j} \longleftrightarrow \sin a\omega$$

Example 5.4: $y(t) = x(t) * \delta(-bt+c)$

Solution: $y(t) = Y(\omega)$

$$y(t) = x(t) * \frac{1}{b}\delta\left(t - \frac{c}{b}\right)$$

$$y(t) = \frac{1}{b} x(t) * \delta\left(t - \frac{c}{b}\right)$$

$$y(t) = \frac{1}{b} x\left(t - \frac{c}{b}\right)$$

$$x(t) \longleftrightarrow X(\omega)$$

$$\frac{1}{b} x\left(t - \frac{c}{b}\right) \longleftrightarrow \frac{e^{-j\omega c/b}}{b} X(\omega) = Y(\omega)$$

One-sided exponential:

$$x(t) = e^{-at}u(t) \quad : \quad a > 0$$

$$x(t) = e^{-at}u(t) \longleftrightarrow X(\omega) = \frac{1}{a+j\omega}$$

$$X(\omega) = |X(\omega)| e^{j\angle X(\omega)}$$

$$\begin{cases} |X(\omega)| = \dfrac{1}{\sqrt{a^2+\omega^2}} \\ \angle X(\omega) = -\tan^{-1}\left(\dfrac{\omega}{a}\right) \end{cases}$$

ω	$\lvert x(\omega)\rvert$	$\angle x(\omega)$
$-\infty$	0	$\pi/2$
$-a$	$\frac{1}{a\sqrt{2}}$	$\pi/4$
0	$\frac{1}{a}$	0
a	$\frac{1}{a\sqrt{2}}$	$-\pi/4$
∞	0	$-\pi/2$

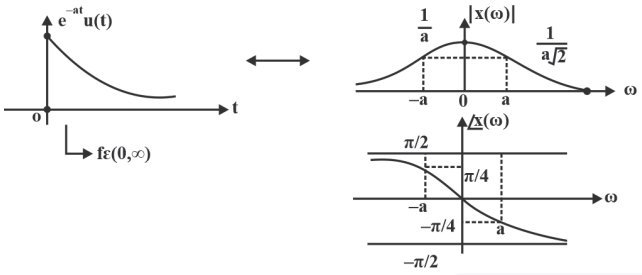


Fig. 5.9

Absolute BW = ∞ Hz

$$3-dB\ B.W. = (a-0)\frac{rad}{\sec}$$

$$e^{-at}u(t) \xrightarrow{3-dB\ B.W.} a\ rad/\sec$$

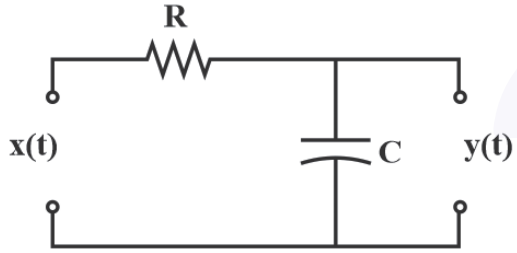


Fig. 5.10

$$H(S) = \frac{1}{1+SRC} \longrightarrow H(\omega) = \frac{1}{1+j\omega RC}$$

$$= \frac{1/RC}{1/RC + j\omega} \longleftrightarrow h(t) = \frac{1}{RC}e^{-t/RC}u(t)$$

$$\frac{1}{RC}\ rad/\sec \qquad 3-dB\ B\omega$$

One-sided exponential:

$$x(t) = e^{at}u(-t) \quad : a > 0$$

$$x(t) = e^{at}u(-t) \longleftrightarrow X(\omega) = \frac{1}{a-j\omega}$$

$$|X(\omega)| = \frac{1}{\sqrt{a^2+\omega^2}}$$

$$\angle X(\omega) = \tan^{-1}\left(\frac{\omega}{a}\right)$$

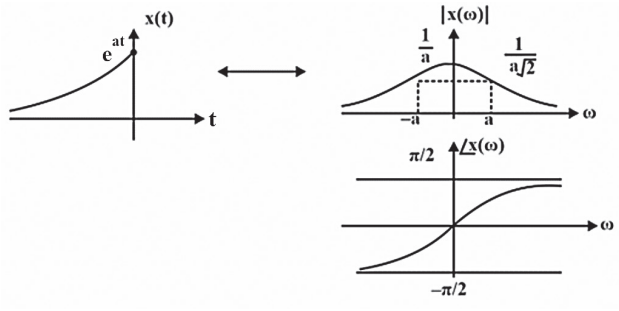


Fig. 5.11

Two-sided exponential:

$$x(t) = e^{-a|t|} \quad : a > 0$$

$$e^{-a|t|} = e^{-at}u(t) + e^{at}u(-t)$$

$$x(t) = e^{-a|t|} \longleftrightarrow X(\omega)$$

$$= \frac{1}{a+j\omega} + \frac{1}{a-j\omega} = \frac{2a}{a^2+\omega^2}$$

$$x(t) = e^{-a|t|} \longleftrightarrow X(\omega) = \frac{2a}{a^2+\omega^2}$$

Two-sided exponential:

$$x(t) = e^{-a|t|}sgn(t) \quad : a > 0$$

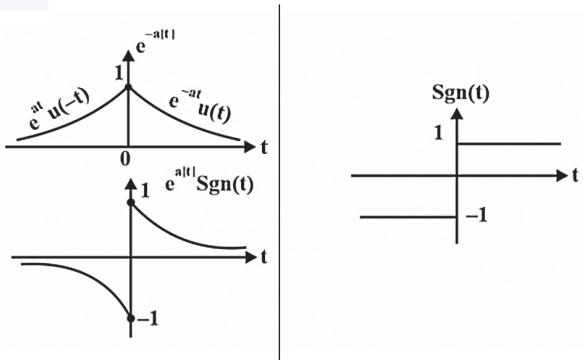


Fig. 5.12

$$e^{-a|t|}\,\mathrm{sgn}(t) = e^{-at}u(t) - e^{at}u(-t)$$

$$e^{-a|t|}\,\mathrm{sgn}(t) \longleftrightarrow \frac{1}{a+j\omega} - \frac{1}{a-j\omega} = \frac{-2j\omega}{a^2+\omega^2}$$

Pair:

$$\delta(t) \longleftrightarrow 1$$

$$e^{-at}u(t) \longleftrightarrow \frac{1}{a+j\omega}$$

$$e^{at}u(-t) \longleftrightarrow \frac{1}{a-j\omega}$$

$$e^{-a|t|} \longleftrightarrow \frac{2a}{a^2+\omega^2}$$

$$e^{-a|t|}\,\mathrm{sgn}(t) \longleftrightarrow \frac{-2j\omega}{a^2+\omega^2}$$

Multiplication with t property, or differentiation in frequency domain:

$$x(t) \longleftrightarrow X(\omega) \qquad x(t) \longleftrightarrow X(f)$$

$$tx(t) \longleftrightarrow \qquad tx(t) \longleftrightarrow$$

$$+j\frac{d}{d\omega}X(\omega) \qquad \frac{+j}{2\pi}\frac{dX(f)}{df}$$

$$t^2x(t) \longleftrightarrow \qquad t^2x(t) \longleftrightarrow$$

$$+j\frac{d}{d\omega}\left\{j\frac{dX(\omega)}{d\omega}\right\} \qquad \frac{+j}{2\pi}\frac{d}{df}\left\{\frac{j}{2\pi}\frac{dX(f)}{df}\right\}$$

$$e^{-at}u(t) \longleftrightarrow \frac{1}{a+j\omega}$$

$$t^ne^{-at}u(t) \longleftrightarrow \frac{n!}{(a+j\omega)^{n+1}}$$

$$t^3e^{-at}u(t) \longleftrightarrow \frac{3!}{(a+j\omega)^4}$$

Example 5.5: $X(f)=\dfrac{1}{1+jf} \overset{IFT}{\longleftrightarrow} x(t)=?$

Solution:

$$X(\omega)=\frac{1}{1+j\dfrac{\omega}{2\pi}}$$

$$X(\omega)=\underbrace{\frac{2\pi}{2\pi+j\omega}}_{a} \longleftrightarrow x(t)=2\pi e^{-2\pi t}u(t)$$

Rectangular or gate pulse:

1. Symmetrical_

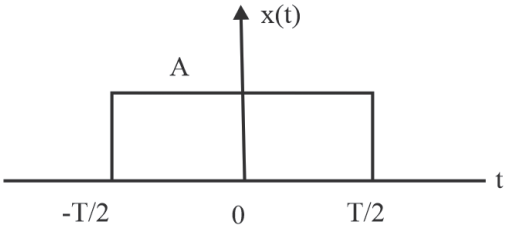


Fig. 5.13

$$\longleftrightarrow X(\omega)=ATSa\left[\frac{\omega T}{2}\right]$$

Zero crossover frequency: $\dfrac{\omega T}{2}=K\pi$

$K \in I, K \neq 0$

$\omega=(2\pi/T)^K$

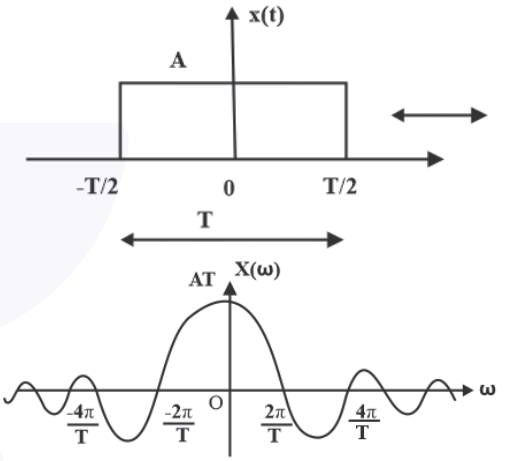


Fig. 5.14

Zero crossover frequency:

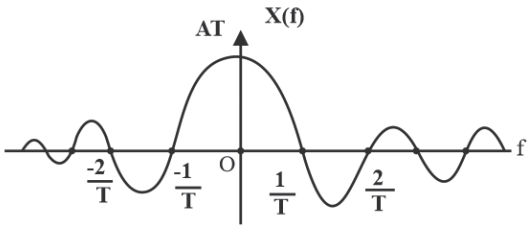


Fig. 5.15

$f = k / duration$

$f = k/T \quad K \in I, K \neq 0$

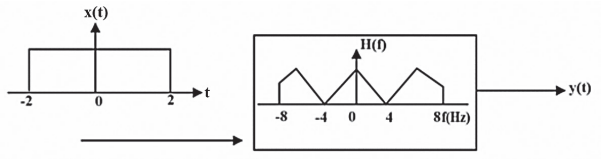


Fig. 5.16

Example 5.6: The absent frequency range and absent frequency components in y(t) = ?

Solution:

y(t): Range → f > 8

Component:

$$x(t) \xrightarrow{\text{Absent components}} f = \frac{K}{4}$$

$K \in I$,

$K \neq 0$

$$f = \frac{1}{4}, \frac{2}{4}, \frac{3}{4}, 1 \ldots\ldots\ldots\ldots (4) \ldots\ldots\ldots\ldots$$

2. Non-symmetrical:_

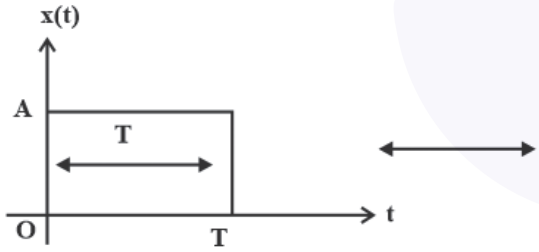


Fig. 5.17

a) $x(t) = Au(t) - Au(t-T) \longleftrightarrow$

$$X(s) = \frac{A}{s}\left(1 - e^{-ST}\right)$$

↓s = jω

$$\boxed{X(\omega) = \frac{A}{j\omega}\left(1 - e^{-j\omega T}\right)}$$

b)

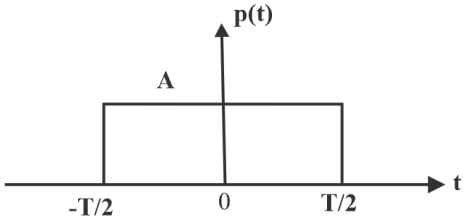


Fig. 5.18

$$\longleftrightarrow P(\omega) = ATSa\left(\frac{\omega T}{2}\right)$$

$$x(t) = p(t - 1/2)$$

$X(\omega)$

$$X(\omega) = e^{-j\frac{\omega T}{2}} p(\omega)$$

$$X(\omega) = e^{-j\omega T/2} ATSa\left(\frac{\omega T}{2}\right)$$

Triangular pulse:

1. Symmetry

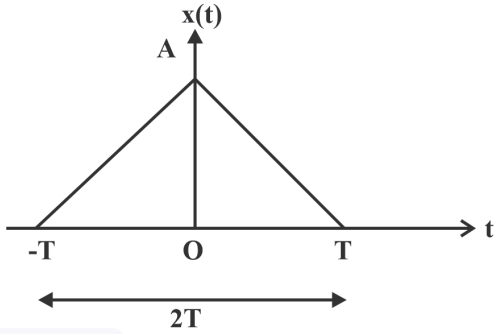


Fig. 5.19

$$\xleftrightarrow{FoTo} X(\omega) = ATSa^2\left[\frac{\omega T}{2}\right]$$

↓

Absent frequency

$$f = \frac{K}{(duration/2)} Hz \qquad K \in I, \; k \neq 0$$

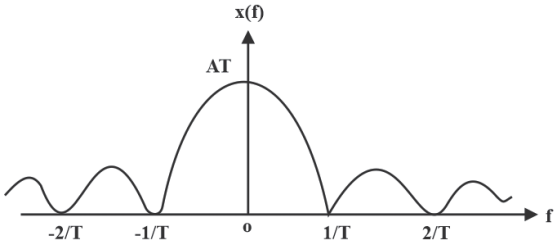


Fig. 5.20

2. Asymmetrical

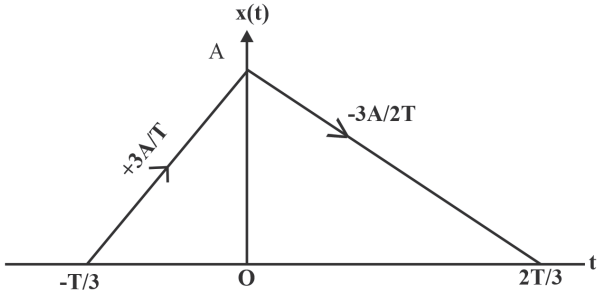


Fig. 5.21

$$X(t)=\frac{3A}{T}r\left(t+\frac{T}{3}\right)-\frac{9A}{2T}r(T)$$
$$+\frac{3A}{2T}r\left(t-\frac{2T}{3}\right)$$

$$X(s)=\frac{3A}{Ts^2}e^{\frac{ST}{3}}-\frac{9A}{2Ts^2}+\frac{3Ae^{-S2T/3}}{2Ts^2}$$

$$x(\omega)=X(s)\Big|_{s=j\omega}$$

Average value of signal:

$$\overline{x(t)}=\underset{T\to\infty}{Lim}\frac{1}{T}\int_{-T/2}^{T/2}x(t)dt$$

T: Symmetrical duration.

Full DC:

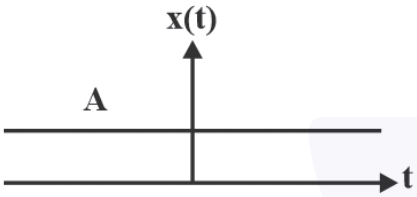


Fig. 5.22

$$\bar{x}(t)=A$$

Half DC:

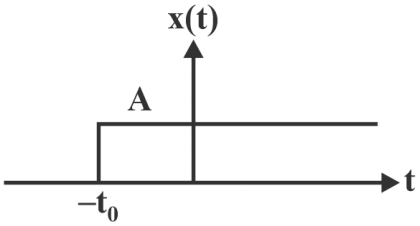


Fig. 5.23

$$\bar{x}(t)=A/2$$

Finite duration finite amplitude:

$$\bar{x}(t)=0$$

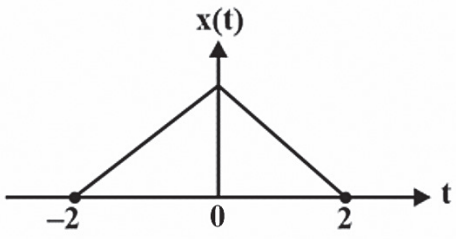


Fig. 5.24

4.

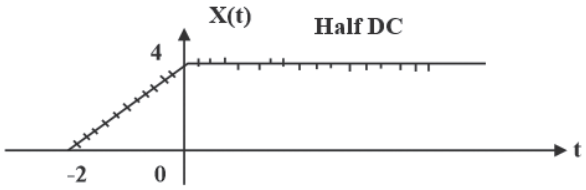


Fig. 5.25

Solution:

$$\bar{x}(t)=0+\frac{4}{2}=2$$

Differentiation property: Valid only when $\bar{x}(t)=0$

$$x(t)\longleftrightarrow X(\omega)\qquad x(t)\longleftrightarrow X(f)$$

$$\frac{dx(t)}{dt}\longleftrightarrow j\omega X(\omega)\quad \frac{dx(t)}{dt}\longleftrightarrow j2\pi fX(f)$$

$$\frac{d^2x(t)}{dt}\longleftrightarrow(j\omega)^2X(\omega)$$

$$\frac{d^2x(t)}{dt^2}\longleftrightarrow(j2\pi f)^2X(f)$$

Problem solving approach:

$$x(t)\longleftrightarrow X(\omega)=?$$

S.1 Calculate $\bar{x}(t)=k\neq 0$

S.2 $p(t)=\frac{dx(t)}{dt}\longleftrightarrow p(\omega)=j\omega X(\omega)$

$$X(\omega)=\frac{P(\omega)}{j\omega}$$

S.3 Final result: $X(\omega)=\frac{P(\omega)}{j\omega}+F.T.\{K\}$

FT of power signals:

$$X(\omega)=\frac{2a}{a^2+\omega^2}$$

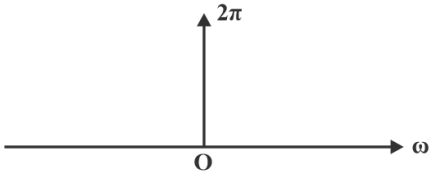
1. $\int_{-\infty}^{\infty}X(\omega)d\omega=\int_{-\infty}^{\infty}\frac{2a}{a^2+\omega^2}d\omega$

$$=\frac{2a}{a}\left(\tan t\frac{\omega}{a}\right)_{-\infty}^{\infty}=2\pi$$

2. $\lim_{a\to 0}\frac{2a}{a^2+\omega^2}=\begin{cases}\infty & :\omega=0\\ 0 & :\omega\neq 0\end{cases}$

$X(\omega)=\frac{2a}{a^2+\omega^2}$

$G(\omega)=\lim_{a\to 0}X(\omega)=2\pi\delta(\omega)$



a) $x(t)=1$: DC signal:

m.1 x(t) = 1:

$e^{-a|t|}\longleftrightarrow\frac{2a}{2^2+\omega^2}$

$\lim_{a\to 0}e^{-a1+1}\longleftrightarrow\lim_{a\to 0}\frac{2a}{a^2+\omega^2}$

$1\longleftrightarrow 2\pi\delta(\omega)$

m.2 *Let* : $X(\omega)=\delta(\omega)$

$x(t)\longleftrightarrow X(\omega)$

IFT: $x(t)=\frac{1}{2\pi}\int_{-\infty}^{\infty}X(\omega)e^{j\omega t}d\omega$

$x(t)=\frac{1}{2\pi}\int_{-\infty}^{\infty}\delta(\omega)e^{j\omega t}d\omega$

b) $x(t)=1/2\pi$

PAIR $\frac{1}{2\pi}\longleftrightarrow\delta(\omega)$ $1\longleftrightarrow 2\pi\delta(\omega)$

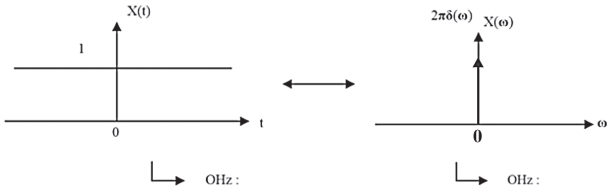


Fig. 5.26

$1\longleftrightarrow 2\pi\delta(\omega)$

$1\longleftrightarrow\delta(f)$

8.

Solution:

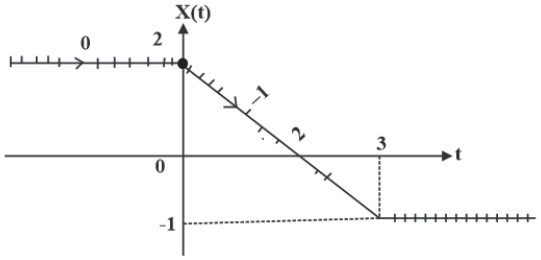


Fig. 5.27

$\longleftrightarrow X(\omega)=?$

$\bar{x}(t)=\frac{2}{2}+0-\frac{1}{2}=\frac{1}{2}$

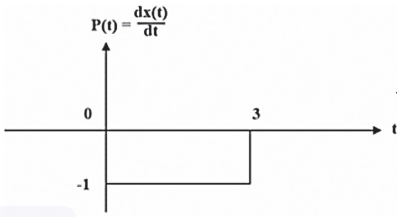


Fig. 5.28

$\longleftrightarrow P(\omega)=j\omega X(\omega)=-e^{-j\frac{3\omega}{2}}3Sa\left(\frac{3\omega}{2}\right)$

$X(\omega)=\frac{-3e^{-j3\omega/2}}{j\omega}Sa\left(\frac{3\omega}{2}\right)+\pi\delta(\omega)$

$Real1\longleftrightarrow 2\pi\delta(\omega)$

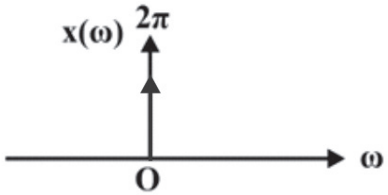


Fig. 5.29

$Complex: e^{j\omega_0 t}1\longleftrightarrow 2\pi\delta(\omega-\omega_0)$

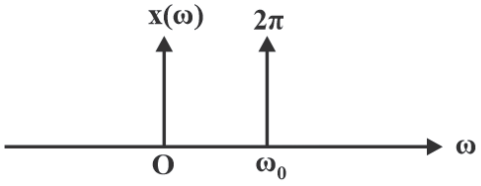


Fig. 5.30

$e^{-j\omega_0 t}\leftrightarrow 2\pi\delta(\omega+\omega_0)$

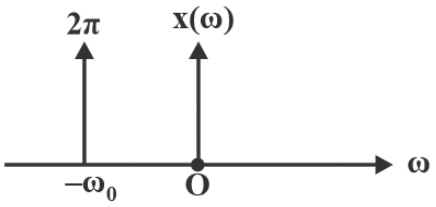


Fig. 5.31

$$x(t) = \cos\omega_0 t \longleftrightarrow \pi\begin{bmatrix}\delta(\omega-\omega_0)\\ +\delta(\omega+\omega_0)\end{bmatrix} = X(\omega)$$

$$\longleftrightarrow \frac{1}{2}\begin{bmatrix}\delta(f-f_0)\\ +\delta(f+f_0)\end{bmatrix} = X(f)$$

$\cos\omega_0 t \longleftrightarrow$

Real :

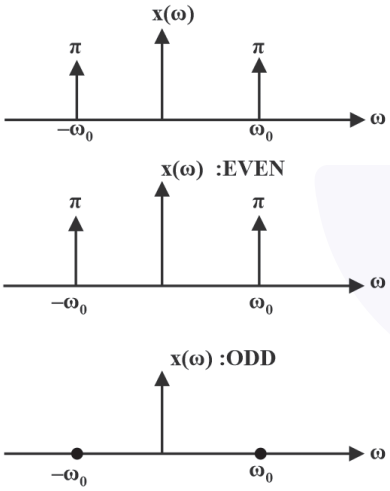


Fig. 5.32

$\cos 2\pi f_0 t \longleftrightarrow$

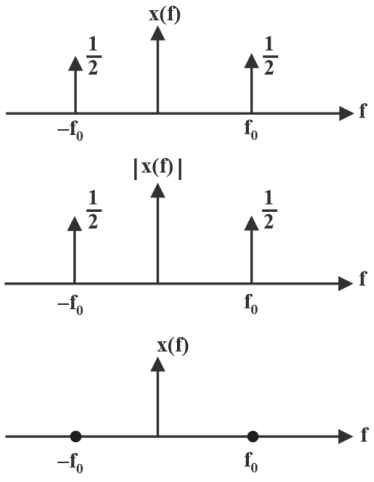


Fig. 5.33

$$\#\sin\omega_0 t \longleftrightarrow \frac{\pi}{j}\delta(\omega-\omega_0) - \frac{\pi}{j}\delta(\omega+\omega_0)$$

$$\underline{\pi\delta(\omega-\omega_0)e^{-j\pi/2}} + \underline{\pi\delta(\omega+\omega_0)e^{j\pi/2}}$$

$$\#\sin 2\pi f_0 t \longleftrightarrow \frac{1}{2j}\delta(f-f_0) - \frac{1}{2j}\delta(f+f_0)$$

$$\underline{\frac{1}{2}\delta(f-f_0)e^{-\frac{j\pi}{2}}} + \underline{\frac{1}{2}\delta(f+f_0)e^{\frac{j\pi}{2}}}$$

$\sin\omega_0 t \longleftrightarrow$

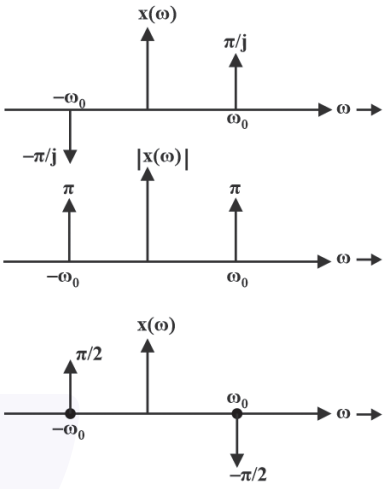


Fig. 5.34

$\sin 2\pi f_0 t \longleftrightarrow$

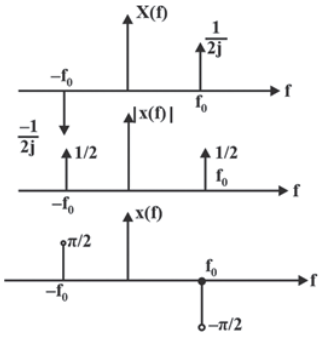


Fig. 5.35

Note: $X(\omega) \longleftrightarrow X(f)$ *same*

$|X(\omega)| \longleftrightarrow |X(f)|$

$\angle X(\omega) \longleftrightarrow \angle X(f)$

only horizontal axis divided by 2π.

1. $x(t) = \text{sgn}(t)$

$$\bar{x}(t) = \frac{-1}{2} + \frac{1}{2} = 0$$

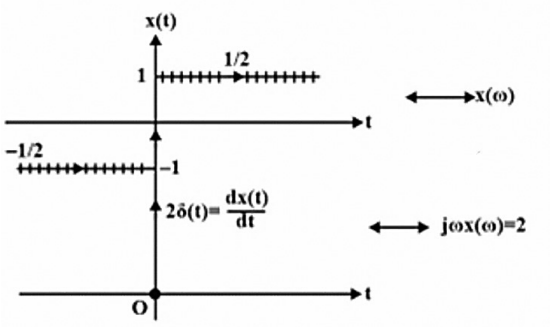


Fig. 5.36

$$X(\omega)=\frac{2}{j\omega}+0$$

2. $\boxed{x(t)=u(t)\longleftrightarrow X(\omega)=\pi\delta(\omega)+\frac{1}{j\omega}}$

Or, $\quad \operatorname{sgn}(t)=2u(t)-1$

$$u(t)=\frac{1}{2}+\frac{1}{2}\operatorname{sgn}(t)$$

$$X(\omega)=\frac{1}{2}\left(2\pi\delta(\omega)\right)+\frac{1}{2}\times\frac{2}{i\omega}$$

$$\boxed{X(\omega)=\pi\delta(\omega)+\frac{1}{j\omega}}$$

Or, $\quad e^{-at}u(t)\longleftrightarrow\frac{1}{a+j\omega}$

$$u(t)=\lim_{a\to 0}e^{-at}u(t)\longleftrightarrow\lim_{a\to 0}\left\{\frac{1}{a+j\omega}\right\}$$

$$\lim_{a\to 0}\left\{\frac{a}{a^2+\omega^2}-\frac{j\omega}{a^2+\omega^2}\right\}$$

$$\lim_{a\to 0}\frac{a}{a^2+\omega^2}-\lim_{a\to 0}\frac{j\omega}{a^2+\omega^2}$$

$$\pi\delta(\omega)+\frac{1}{j\omega}$$

Pair:

$$1\longleftrightarrow 2\pi\delta(\omega)$$

$$\cos\omega_0 t\longleftrightarrow\pi\delta(\omega-\omega_0)+\pi\delta(\omega+\omega_0)$$

$$\sin\omega_0 t\longleftrightarrow\frac{\pi}{j}\delta(\omega-\omega_0)-\frac{\pi}{j}\delta(\omega+\omega_0)$$

$$u(t)\longleftrightarrow\pi\delta(\omega)+\frac{1}{j\omega}$$

$$\operatorname{sgn}(t)\longleftrightarrow\frac{2}{j\omega}$$

Duality:

$$x(t)\longleftrightarrow X(\omega)\qquad x(t)\longleftrightarrow X(f)$$

$$X(t)\longleftrightarrow 2\pi x(-\omega)\qquad X(t)\longleftrightarrow x(-f)$$

1. $u(t)\longleftrightarrow\pi\delta(\omega)+\frac{1}{j\omega}$

$$\pi\delta(t)+\frac{1}{jt}\longleftrightarrow 2\pi u(-\omega)$$

2. $\operatorname{sgn}(t)\longleftrightarrow\frac{2}{j\omega}$

$$\frac{2}{jt}\longleftrightarrow 2\pi\operatorname{sgn}(-\omega)$$

3. $e^{-a|t|}\longleftrightarrow\frac{2a}{a^2+\omega^2}$

$$\frac{2a}{a^2+t^2}\longleftrightarrow 2\pi e^{-a|-\omega|}$$

Purpose: $x(t)\longleftrightarrow X(\omega)$

S.1 Try to find FT of x (t):

S.2 If S.1 fails.

$$x(t)\big|_{t=\omega}=G(\omega)$$

S.3 $g(t)\longleftrightarrow G(\omega)$: $\quad[\textit{known Transforms}]$

$$G(t)\longleftrightarrow 2\pi g(-\omega)$$

9. $x(t)=\frac{1}{a+jt}\longleftrightarrow X(\omega)$

Solution: $\rightarrow\quad x(t)\big|_{t=\omega}: G(\omega)=\frac{1}{a+j\omega}$

$$\rightarrow\quad e^{-at}u(t)\longleftrightarrow\frac{1}{a+j\omega}$$

$$\frac{1}{a+jt}\longleftrightarrow 2\pi e^{-a(-\omega)}u(-\omega)=2\pi e^{a\omega}u(-\omega)$$

Conclusions:

1. $1\cdot\left(\frac{\sin at}{\pi t}\right)\overset{F.T.}{\longleftrightarrow}$

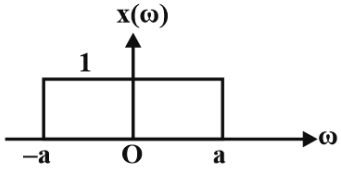


Fig. 5.37

2. $1 \cdot \left(\frac{\sin at}{\pi t}\right)^2 \xleftrightarrow{F.T.}$

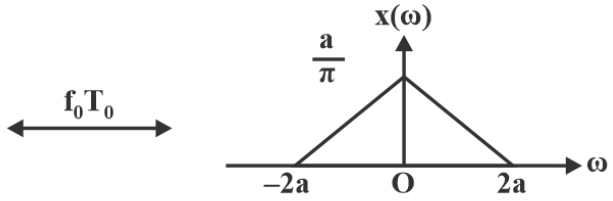


Fig. 5.38

Important results:

1.

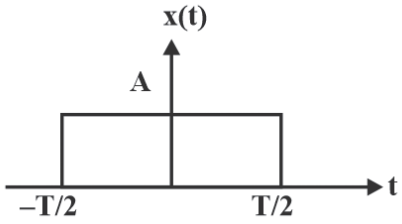


Fig. 5.39

$x(\omega) = ATSa(\omega T / 2)$

$X(0) = \int_{-\infty}^{\infty} x(t)dt$

2.

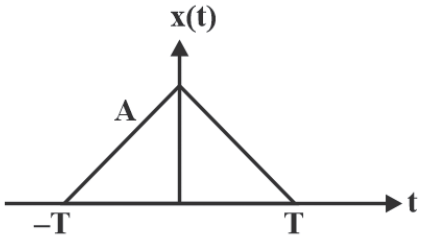


Fig. 5.40

$X(\omega) = ATSa^2\left[\frac{\omega T}{2}\right]$

3. $\frac{\sin at}{\pi t}$

$\sin c(t) or\ sa(t)$

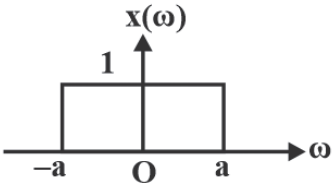


Fig. 5.41

Sin c (t) and sa(t) differences should be clearly explained, and check the above equation as per the definition.

4. $\left(\frac{\sin at}{\pi t}\right)^2$

$\sin C^2(t), Sa^2(t)$

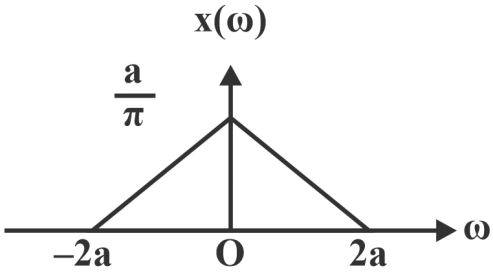


Fig. 5.42

5. $X(\omega) = Sa(\omega) \longleftrightarrow x(t)$

Sa(ω) or $\frac{\sin \omega}{\omega}$

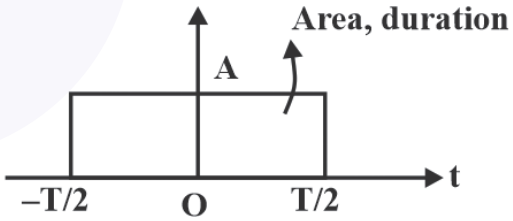


Fig. 5.43

6. $x(\omega) = Sa^2(\omega)$

$\sin c^2(\omega)$

$(\sin \omega / \omega)^2$

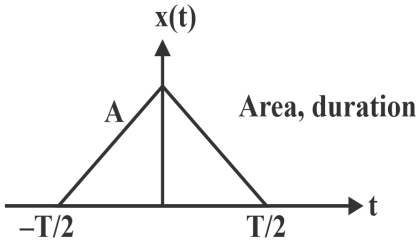


Fig. 5.44

7.

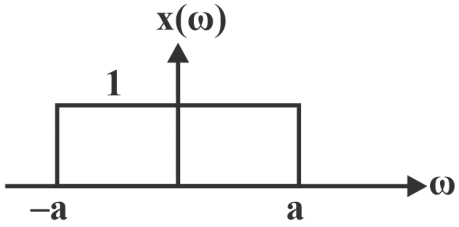


Fig. 5.45

$$x(t) = \frac{\sin at}{\pi t}$$

8.

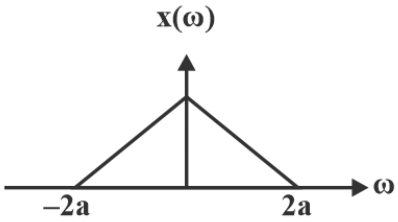


Fig. 5.46

$$x(t) = \left(\frac{\sin at}{\pi t}\right)^2$$

10. $x(t) = sa(t) \longleftrightarrow X(\omega)$

Solution:

$$x(t) = \frac{\sin t}{t}$$

$$x(t) = \pi\left\{\frac{\sin t}{\pi t}\right\}$$

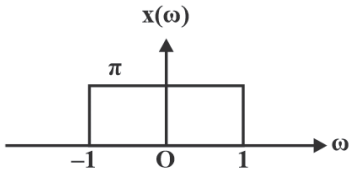


Fig. 5.47

Area Under the Signal Property

1. $X(\omega) = \int_{-\infty}^{\infty} x(t)e^{-j\omega t}dt$

2. $x(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(\omega)e^{j\omega t}d\omega$

$$X(0) = \int_{-\infty}^{\infty} x(t)dt \qquad = \int_{-\infty}^{\infty} X(f)e^{j2\pi ft}df$$

$$x(0) = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(\omega)d\omega = \int_{-\infty}^{\infty} X(f)df$$

Problem:

1. $\int_{-\infty}^{\infty} \underline{x(t)}dt = \mathbf{X}(\omega)\big|_{\omega=0}$

2. $\int_{-\infty}^{\infty} \underline{x(f)}df = x(t)\big|_{t=0}$

3. $\int_{-\infty}^{\infty} \underline{X(\omega)}d\omega = 2x\pi(t)\big|_{t=0}$

11. $\int_{-\infty}^{\infty} \text{sinc}(t)dt$

Solution:

$$\text{sinc}(t) = \left\{\frac{\sin \pi t}{\pi t}\right\}$$

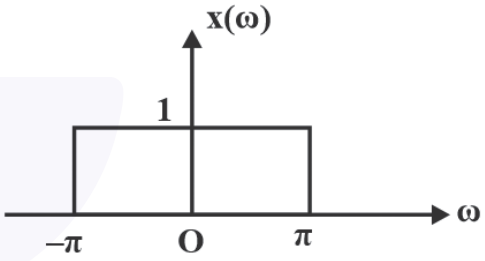


Fig. 5.48

$$\int_{-\infty}^{\infty} \text{sinc}(t)dt = 1$$

$$\downarrow$$

$$x(t) \longleftrightarrow X(\omega)\big|_{\omega=0}$$

12. $\int_{-\infty}^{\infty} \underbrace{e^{-at}u(t)}_{a>0}dt = X(\omega)\Big|_{\omega=0} = \frac{1}{a}$

13. $\int_{-\infty}^{\infty} \underbrace{e^{a|t|}}_{a>0}dt = X(\omega)\Big|_{\omega=0} = \left\{\frac{2a}{a^2+\omega^2}\right\}_{\omega=0} = 2/a$

14. $\int_{-\infty}^{\infty} \underbrace{\frac{1}{a+j\omega}}_{X(\omega)}d\omega = 2\pi x(0)$

$$= 2\pi\left(e^{-at}u(t)\right)_{t=0} = 2\pi$$

Convolution property:

$$x_1(t) \longleftrightarrow X_1(\omega) \qquad x_1(t) \longleftrightarrow X_1(f)$$

$$x_2(t) \longleftrightarrow X_2(\omega) \qquad x_2(t) \longleftrightarrow X_2(f)$$

Purpose $x(t) = x_1(t) * x_2(t) = ?$

Method 1 : $x(t) = \int_{-\infty}^{\infty} x_1(\tau)\, x_2(t-\tau)\, d\tau$

Method 2 : $x(t) = x_1(t) * x_2(t) \longleftrightarrow X(\omega) = X_1(\omega)\, X_2(\omega)$

S.1 calculate X(ω) X(f) = X_1(f) X_2 (f)

S.2 $X(\omega) \xleftrightarrow{IFT} x(t)$

Conclusion:

$$\text{sinc}(\overset{\swarrow}{4t}) * \sin(\overset{\swarrow}{6t}) = \frac{1}{6}\text{sinc}(4t)$$

$$A\,\text{sinc}(\alpha t) * B\,\text{sinc}(\beta t) = AB\frac{[1\text{sinc}[pt]]}{K}$$

$K = \max(\alpha, \beta)$

$p = \min(\alpha, \beta)$

15. sin c(t) * sin c(t) * sin c(t) * ... 50,000 times.

Solution:

= sin c (t)

Multiplication in time domain:

$x_1(t) \longleftrightarrow X_1(\omega)$ $x_1(t) \longleftrightarrow X_1(f)$

$x_2(t) \longleftrightarrow X_2(\omega)$ $x_2(t) \longleftrightarrow X_2(f)$

$$x_1(t).x_2(t) \longleftrightarrow \frac{1}{2\pi}[X_1(\omega) * X_2(\omega)]$$

$$x_1(t).x_2(t) \longleftrightarrow X_1(f) * X_2(f)$$

$$\longrightarrow \frac{1}{2\pi}\int_{-\infty}^{\infty} X_1(\lambda)\, X_2(\omega-\lambda)\, d\lambda$$

$$\longrightarrow \int_{-\infty}^{\infty} X_1(\lambda)\, X_2(\text{f}-\lambda)\, d\lambda$$

$$x_1(t) * x_2(t) \longleftrightarrow X_1(\omega)\, X_2(\omega)$$

$$x_1(t) * x_2(t) \longleftrightarrow X_1(f)\, X_2(f)$$

$$X_1(\omega) * X_2(\omega) \longleftrightarrow 2\pi(x_1(t)\, x_2(t))$$

$$X_1(f) * X_2(f) \longleftrightarrow x_1(t).x_2(t)$$

$$A\,\text{sinc}(\alpha\omega) * B\,\text{sinc}(\beta\omega) = \left\{AB \times \frac{1}{k}\text{sinc}[p\omega]\right\}$$

$k = \max(\alpha, \beta)$

$p = \min(\alpha, \beta)$

Note:

$$A\,\text{sinc}(\alpha t) * B\,\text{sinc}(\beta t) = AB\left\{\frac{1}{k}\text{sinc}[pt]\right\}$$

$k = \max(\alpha, \beta)$

$p = \min(\alpha, \beta)$

$$A\,\text{sinc}(\alpha\omega) * B\,\text{sinc}(\beta\omega) = AB\left\{\frac{1}{k}\text{sinc}[p\omega]\right\}$$

$k = \max(\alpha, \beta)$

$p = \min(\alpha, \beta)$

Integration property:

$x(t) \longleftrightarrow X(\omega)$

$$\int_{-\infty}^{t} x(\tau)\, d\tau = x(t) * u(t) \longleftrightarrow$$

$$X(\omega) \cdot \left\{\pi\delta(\omega) + \frac{1}{j\omega}\right\}$$

$$\frac{X(\omega)}{j\omega} + \pi\delta(\omega)\, X(\omega)$$

$$\frac{X(\omega)}{j\omega} + \underline{\pi X(0)\,\delta(\omega)}$$

$$\int_{-\infty}^{t} x(\tau)\, d\tau \longleftrightarrow \frac{X(\omega)}{j\omega} + \pi\delta(\omega)\, X(0)$$

$$\int_{-\infty}^{t} x(\tau)\, d\tau \longleftrightarrow \frac{X(f)}{j2\pi f} + \frac{1}{2}\delta(f)\, X(0)$$

Complex conjugate property:

$x(t) \longleftrightarrow X(\omega)$ $x(t) \longleftrightarrow X(f)$

$x^*(t) \longleftrightarrow X^*(-\omega)$ $x^*(t) \longleftrightarrow X^*(-f)$

$x(t)$	$X(\omega)$
E	E
0	0
NENO	*NENO*

$x(t)$	$X(\omega)$
Real	C.S.
Imaginary	C.A.S.
C.S.	Real
C.A.S	Imaginary

$x(t)$	$X(\omega)$
RE	RE
RO	IO
IE	IE
IO	RO

16. $x(t)$ *is* Real:

Solution:

$x(t) = x^*(t)$

$X(\omega) = X^*(-\omega) \rightarrow$ C.S.

Important points:

1. $x(t)$: Real : $\begin{cases} X(\omega) \text{ is conjugate symmetry} \\ |X(\omega)| \text{ vs } \omega \text{ is EVEN} \\ \angle X(\omega) \text{ vs } \omega \text{ is ODD} \end{cases}$

2. $x(t)$: Imaginary : $\begin{cases} X(\omega) \text{ is conjugate} \\ \text{antisymmetric} \\ |X(\omega)| \text{ vs } \omega \text{ is EVEN} \\ \angle X(\omega) \text{ vs } \omega \text{ is NENO.} \end{cases}$

Note: If the magnitude spectrum is not even. $x(t)$: complex

17.

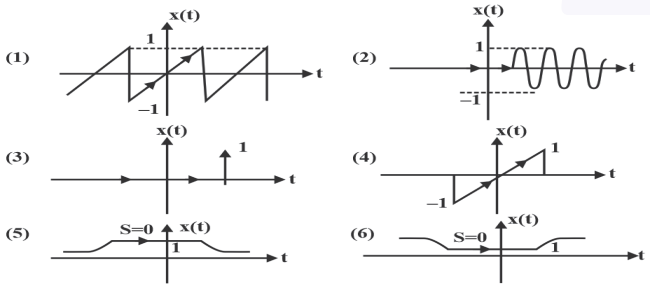


Fig. 5.49

Solution:

1. $\text{Re}\{X(\omega)\} = 0$

$X(\omega) = X_R(\omega) + jX_I(\omega) \longleftrightarrow$

$x(t)$: C.A.S.

$x(\omega) = I + E \longleftrightarrow x(t) : I + E$

$I + O \longleftrightarrow x(t) : R + O$

2. $\text{Im}\{X(\omega)\} = 0$

$X(\omega)$: Real $\longleftrightarrow x(t)$: C.S.

$X(\omega) \qquad x(t)$

$\text{Real} + E \longrightarrow R + E$

$\text{Real} + O \longrightarrow I + O$

3. $\int_{-\infty}^{\infty} X(\omega)\,d\omega = 0$

$\int_{-\infty}^{\infty} X(\omega)\,d\omega = 2\pi x(0) = 0$

$x(t)\big|_{t=0} = 0$

4. $\int_{-\infty}^{\infty} \underbrace{\left\{\frac{dX(\omega)}{d\omega}\right\}}\, d\omega = 0$

$\int_{-\infty}^{\infty} p(\omega)\,d\omega = 2\pi p(t)\big|_{t=0} = 2\pi p(0) = 0$

$p(0) = 0$

$\rho(\omega) = \frac{dX(\omega)}{d\omega}$

$x(t) \longleftrightarrow X(\omega)$

$tx(t) \longleftrightarrow j\frac{dX(\omega)}{d\omega}$

$p(t) = -jtx(t) \longleftrightarrow \frac{dX(\omega)}{d\omega} = \rho(\omega p)$

5. $\int_{-\infty}^{\infty} \underbrace{\omega X(\omega)}_{p(\omega)}\, d\omega = 0$

$\int_{-\infty}^{\infty} p(\omega)\,d\omega = 2\pi p(0) = 0 \longrightarrow p(0) = 0$

$x(t) \longleftrightarrow X(\omega)$

$-j\frac{dx(t)}{dt} \longleftrightarrow \omega X(\omega)$

$p(t) = -j\frac{dx(t)}{dt}$

$(p(t))_{t=0} = -j\left(\frac{dx(t)}{dt}\right)_{t=0} = 0$

Parseval's Energy Theorem

1. Used to calculate specific integration:

$$\int_{-\infty}^{\infty} \underbrace{\left(x_1(t)\,x_2(t)\right)}_{x(t)} dt$$

a) $X(\omega)\big|_{\omega=0}$

b) Parseval theorem.

2. Used to calculate the energy of x (t).

Case a) $\int_{-\infty}^{\infty} x_1(t)\,x_2(t)\,dt$

$$= \int_{-\infty}^{\infty} x_1(t) \times \frac{1}{2\pi}\left\{\int_{-\infty}^{\infty} X_2(\omega) e^{j\omega t} d\omega\right\} dt$$

$$X_1(\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt$$

$$= \frac{1}{2\pi}\int_{-\infty}^{\infty} X_2(\omega)\left\{\int_{-\infty}^{\infty} x_1(t) e^{j\omega t} dt\right\} d\omega$$

$$= \frac{1}{2\pi}\int_{-\infty}^{\infty} X_2(\omega) X_1(-\omega)\, d\omega$$

$$= \frac{1}{2\pi}\int_{-\infty}^{\infty} X_1(\omega) X_2(-\omega)\, d\omega$$

$$\int_{-\infty}^{\infty} x_1(t)x_2(t)dt = \frac{1}{2\pi}\int_{-\infty}^{\infty} X_1(\omega) X_2(-\omega)\, d\omega$$

$$= \frac{1}{2\pi}\int_{-\infty}^{\infty} X_1(-\omega) X_2(\omega)\, d\omega$$

$$= \int_{-\infty}^{\infty} X_1(f) X_2(-f)\, df = \int_{-\infty}^{\infty} X_1(-f) X_2(f)\, df$$

Case b) $\int_{-\infty}^{\infty} x^2(t)\,dt = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(\omega) X(-\omega)\, d\omega$

$$= \int_{-\infty}^{\infty} X(f) X(-f)\, df$$

Case c)

$$\int_{-\infty}^{\infty} x_1(t)\, x_2{}^*(t)\, dt$$

$$= \int_{-\infty}^{\infty} x_1(t)\left\{\frac{1}{2\pi}\int_{-\infty}^{\infty} X_2(\omega) e^{j\omega t} d\omega\right\}^* dt$$

$$= \frac{1}{2\pi}\int_{-\infty}^{\infty} x_1(t)\left\{\int_{-\infty}^{\infty} X_2{}^*(\omega) e^{-j\omega t} d\omega\right\} dt$$

$$= \frac{1}{2\pi}\int_{-\infty}^{\infty} X_2{}^*(\omega)\left\{\int_{-\infty}^{\infty} x_1(t) \overset{X_1(\omega)}{\overbrace{e^{-j\omega t}}}\, dt\right\} d\omega$$

3. $\int_{-\infty}^{\infty} x_1(t)\, x_2{}^*(t)\, dt = \frac{1}{2\pi}\int_{-\infty}^{\infty} X_1(\omega) X_2{}^*(\omega)\, d\omega$

$$= \int_{-\infty}^{\infty} X_1(f) X_2{}^*(f)\, df$$

4. $\int_{-\infty}^{\infty} x(t).x^*(t)\,dt = \int_{-\infty}^{\infty} |x(t)|^2\, dt = E_x$

$$\int_{-\infty}^{\infty} x(t).x^*(t)\,dt$$

$$= \frac{1}{2\pi}\int_{-\infty}^{\infty} X(\omega) X^*(\omega)\, d\omega = \frac{1}{2\pi}\int_{-\infty}^{\infty} |X(\omega)|^2\, d\omega$$

$$= \int_{-\infty}^{\infty} |X(f)|^2 df$$

$x(t) \longleftrightarrow X(\omega)$ $\qquad$ $x(t) \longleftrightarrow X(f)$

$$E_x = \int_{-\infty}^{\infty} |x(t)|^2\, dt = \frac{1}{2\pi}\int_{-\infty}^{\infty} |X(\omega)|^2\, d\omega$$

$$E_x = \int_{-\infty}^{\infty} |x(t)|^2\, dt = \int_{-\infty}^{\infty} |X(f)|^2\, df$$

$|X(\omega)|^2 = G_x(\omega) = E.S.D.$

$|X(f)|^2 = G_x(f) = E.S.D.$

$$E_x = \frac{1}{2\pi}\int_{-\infty}^{\infty} G_x(\omega)\, d\omega$$

$$E_x = \int_{-\infty}^{\infty} G_x(f)\, df$$

Note: $x(t) \longleftrightarrow X(\omega)$

$$\frac{1}{2\pi}\int_{-\omega_c}^{\omega_c} |X(\omega)|^2\, d\omega = m\%\left\{\int_{-\infty}^{\infty} |x(t)|^2\, dt\right\}$$

$x(t) \longleftrightarrow X(f)$

$$\int_{-\omega_c}^{\omega_c} |X(f)|^2\, df = m\%\left\{\int_{-\infty}^{\infty} |x(t)|^2\, dt\right\}$$

Fourier transform of Gaussian pulse:

18. $x(t) = e^{-t^2} \longleftrightarrow X(\omega) = ?$

Solution:

$x(t) = e^{-t^2}$: Zero mean Gaussian pulse:

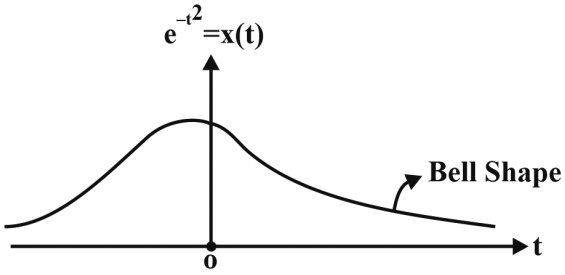


Fig. 5.50

$$FT\left[\frac{dx(t)}{dt}\right] = j\omega X(\omega)$$

$$FT[tx(t)] = +\frac{jdX(\omega)}{d\omega}$$

$$X(t) = e^{-t^2} \longleftrightarrow X(\omega)$$

$$FT\left[\frac{d\left(e^{-t^2}\right)}{dt}\right] = j\omega X(\omega)$$

$$FT\left[-2t\underbrace{e^{-t^2}}_{x(t)}\right] = j\omega X(\omega)$$

$$FT[tx(t)] = \frac{-j\omega}{2}X(\omega)$$

$$+j\frac{d\times(\omega)}{d\omega} = \frac{-j\omega}{2}X(\omega)$$

$$\frac{dX(\omega)}{X(\omega)} = -\frac{\omega}{2}d\omega$$

$$\ln\{X(\omega)\} = -\frac{\omega^2}{4} + c$$

$$X(\omega) = e^{-\frac{\omega^2}{4}}e^{c}$$

$$X(\omega) = ke^{\frac{-\omega^2}{4}}$$

$$X(\omega)\Big|_{\omega=0} = K$$

$$X(\omega)_{\omega=0} = \int_{-\infty}^{\infty} e^{-t^2}dt = \sqrt{\pi}$$

$$k = \sqrt{\pi}$$

$$e^{-t^2} \longleftrightarrow \sqrt{\pi}e^{\frac{-\omega^2}{4}}$$

1. $e^{-at^2} \longleftrightarrow \sqrt{\frac{\pi}{a}}e^{\frac{-\omega^2}{4a}}$

2. $e^{-\pi t^2} \longleftrightarrow e^{\frac{-\omega^2}{4\pi}}$

3. $e^{-\pi t^2} \longleftrightarrow e^{-\pi f^2}$

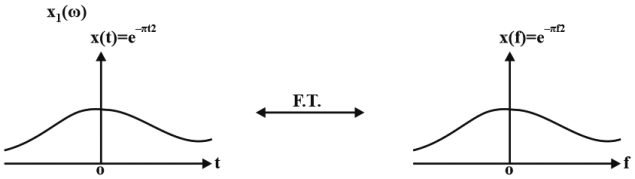


Fig. 5.51

Fourier transform of periodic signals:

1. Let x (t) be a periodic signal with time period T_0 , $x(t) \xleftrightarrow{F.S.C.} C_n$

2. $x(t) = \sum_{n=-\infty}^{n=\infty} C_n e^{jn\omega_0 t}$

$$1 \longleftrightarrow 2\pi\delta(\omega)$$

$$e^{jn\omega_0 t} \longleftrightarrow 2\pi\delta(\omega - n\omega_0)$$

$$X(\omega) = \sum_{n=-\infty}^{\infty} C_n 2\pi\delta(\omega - n\omega_0)$$

$$x(t) \longleftrightarrow X(\omega) = 2\pi\sum_{n=-\infty}^{\infty} C_n\delta(\omega - n\omega_0)$$

$$X(f) = \sum_{n=-\infty}^{\infty} C_n\delta(f - nf_0)$$

19.

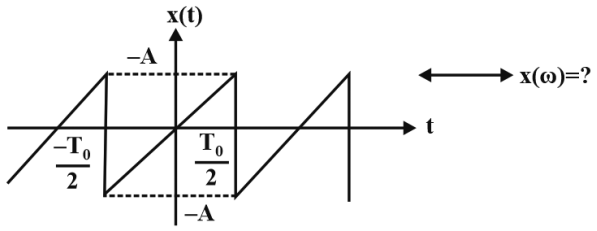


Fig. 5.52

Solution:

S.1 $x(t) \xleftrightarrow{F.S..C.} C_n$

S.2 $X(\omega) = 2\pi \sum_{n=-\infty}^{\infty} C_n \delta(\omega - n\omega_0)$

Characterisation of LTI system:

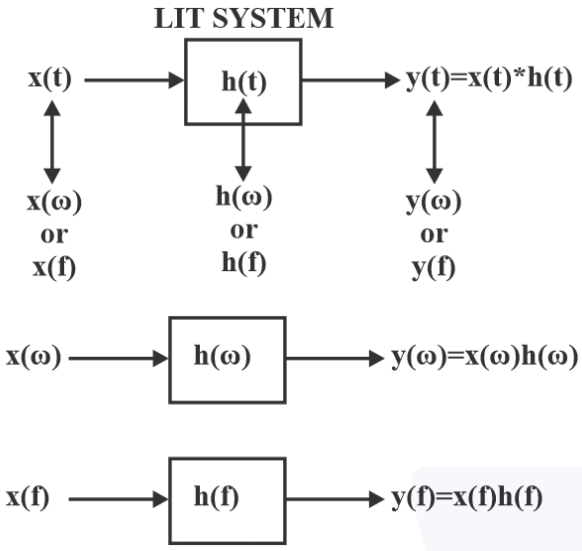


Fig. 5.53

$$\rightarrow Y(\omega) = X(\omega)H(\omega)$$

$$|Y(\omega)|e^{j\angle Y(\omega)} = |X(\omega)|e^{j\angle X(\omega)}|H(\omega)|e^{j\angle H(\omega)}$$

$$|Y(\omega)|e^{j\angle Y(\omega)} = |X(\omega)||H(\omega)|e^{j[\angle X(\omega)+\angle H(\omega)]}$$

$$|Y(\omega)| = |X(\omega)||H(\omega)|$$

$$\angle Y(\omega) = \angle X(\omega) + \angle H(\omega)$$

$$Y(f) = X(f)H(f)$$

$$|Y(f)| = |X(f)||H(f)|$$

$$\angle Y(f) = \angle X(f) + \angle H(f)$$

Frequency response of an LTI system:

1. FT of h(t) is known as the frequency response of the LTI system:
2. $$h(t) \longleftrightarrow H(\omega) = \frac{Y(\omega)}{X(\omega)}$$
$$h(t) \longleftrightarrow H(f) = \frac{Y(f)}{X(f)}$$

Linear and non-linear phase:

1. Consider a continuous-time LTI system with impulse response h(t) and frequency response H(ω).
2. Let a signal x(t) with Fourier transform X(ω) be applied to the input of the system.
3. Let a signal y(t) with Fourier transform Y(ω) denote the output of the system.

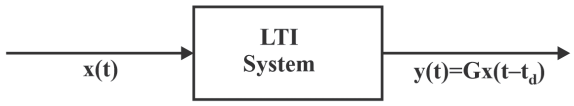


Fig. 5.54

Fig. Time-Domain Condition for Distortion-Less Transmission of a Signal through an LTI System

$$y(t) = Gx(t - t_d)$$

Where the constant G accounts for a change in amplitude and the constant 't_d' accounts for a delay in transmission. Taking the Fourier transform of the above equation yields

$$Y(t) = GX(\omega)e^{-j\omega t_d}$$

$$\frac{Y(\omega)}{X(\omega)} = H(\omega) = Ge^{-j\omega t_d}$$

This is the frequency response required of a system for distortion-less transmission. From this equation, it follows that

a) The magnitude response $|H(\omega)|$ must be a constant, i.e., we must have $|H(\omega)| = G$ For some constant G

b) The phase response $\angle H(\omega)$ must be a linear function of ω with slope $-t_d$ and intercept zero, i.e., we must have $\angle H(\omega) = -\omega t_d$

These two conditions are illustrated in fig (a) and (b), respectively.

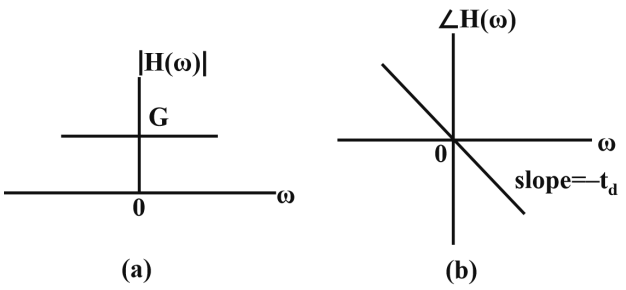


Fig. 5.55 Frequency Response for Distortion-Less Transmission through an LTI System: (a) Magnitude Response and (b) Phase Response

b) The gain $|H(\omega)| = G$ means that every spectral component is multiplied by a constant G.

c) A linear phase $\angle H|(\omega)| = -\omega t_d$ means that every spectral component is delayed by $\mathbf{t_d}$ seconds. For distortion-less transmission, we require a linear phase characteristic.

Phase delay:

1. The time delay experienced by a single-frequency signal (i.e., a sinusoidal signal) when the signal passes through a system is referred to as the system phase delay.
2. Now let us assume that the input signal is the single-frequency signal

$$x(t) = A\cos(\omega_0 t + \theta)$$

$$x(t) = \frac{A}{2}e^{j(\omega_0 t+\theta)} + \frac{A}{2}e^{-j(\omega_0 t+\theta)}$$

And the system frequency response is

$$H(\omega) = |H(\omega)|e^{j\angle H(\omega)}$$

Then, the system output signal is

$$y(t) = \frac{A}{2}|H(\omega_0)|e^{j(\omega_0 t+\theta+\angle H(\omega_0))}$$

$$+\frac{A}{2}|H(-\omega_0)|e^{-j(\omega_0 t+\theta-\angle H(-\omega_0))}$$

Since $|H(\omega_0)| = |H(-\omega_0)| and \angle H(\omega_0)$

$= -\angle H(-\omega_0)$, *we have*

$$y(t) = \frac{A}{2}|H(\omega_0)|e^{j(\omega_0 t+\theta+\angle H(\omega_0))}$$

$$+\frac{A}{2}|H(\omega_0)|e^{-j(\omega_0 t+\theta+\angle H(\omega_0))}$$

$$y(t) = A|H(\omega_0)|\cos(\omega_0 t + \angle H(\omega_0) + \theta)$$

$$= A|H(\omega_0)|\cos\left(\omega_0\left(t + \frac{\angle H(\omega_0)}{\omega_0}\right) + \theta\right)$$

$$y(t) = A|H(\omega_0)|\cos(\omega_0(t - \tau_p(\omega_0)) + \theta)$$

Where,

$$\tau_p(\omega_0) = -\frac{\angle H(\omega_0)}{\omega_0}$$

is the time delay experienced by the single-frequency signal with frequency ω_0 when it passes through the system. Thus, $\tau_p(\omega_0)$ is the system phase delay for a signal with frequency ω_0.

Group delay:

When the input signal contains many sinusoidal components with different frequencies that are not harmonically related, each component will go through different phase delays when processed by a frequency-selective LTI system, and the signal delay is determined using a different parameter called the group delay. $\tau_g(\omega) = -\frac{d\angle H(\omega)}{d\omega}$

20. Consider the following frequency response for a causal and stable LTI system. $H(\omega) = \frac{1-j\omega}{1+j\omega}$

a) Show that $|H(\omega)|$ = A, and determine the value of A.

b) Determine which of the following statements is true about $\tau_g(\omega)$, the group delay.

i. $\tau_g(\omega) = 0, for\ \omega > 0$

ii. $\tau_g(\omega) => 0, for\ \omega > 0$

iii. $\tau_g(\omega) =< 0, for\ \omega > 0$

Solution:

a) Given that

$$H(\omega) = \frac{1-j\omega}{1+j\omega}$$

$$|H(\omega)| = \frac{\sqrt{1+\omega^2}}{\sqrt{1+\omega^2}} = 1$$

Therefore A = 1

b) The phase response of the system is

$$\angle H(\omega) = \tan^{-1}(-\omega) - \tan^{-1}(\omega)$$

$$= -2\tan^{-1}(\omega)$$

Therefore, the group delay is

$$\tau_g(\omega) = -\frac{d\angle(H(\omega))}{d\omega}$$

$$= \frac{d}{d\omega} 2\tan^{-1}(\omega) = \frac{2}{1+\omega^2}$$

Since $\tau_g(\omega) > 0$ for ω>0, statement (ii) is true.

Filtering process: LTIS/S ⟶ Filter

1. $\xrightarrow{x(t)} h(t) \longrightarrow y(t) = x(t) * h(t)$
2. $\xrightarrow{X(\omega)} H(\omega) \longrightarrow Y(\omega) = X(\omega)H(\omega)$
3. $\xrightarrow{|X(\omega)|} |H(\omega)| \longrightarrow |Y(\omega)|$
   $= |X(\omega)||H(\omega)|$
4. $\xrightarrow{\angle x(\omega)} \angle H(\omega) \longrightarrow \angle Y(\omega)$
   $= \angle X(\omega) + \angle H(\omega)$
5. $\xrightarrow{X(f)} H(f) \xrightarrow{Y(f)} = X(f)H(f)$
6. $\xrightarrow{|X(f)|} |H(f)| \xrightarrow{|Y(f)|} = |X(f)||H(f)|$
7. $\xrightarrow{\angle X(f)} \angle H(f) \xrightarrow{\angle Y(f)=} \angle X(f) + \angle H(f)$

Energy calculation:

$X(\omega)$ or $X(f) \longrightarrow H(\omega)$ or $H(f)$

$\longrightarrow Y(\omega)$ or $Y(f)$

1. $E_x = \frac{1}{2\pi}\int_{-\infty}^{\infty} |X(\omega)|^2 d\omega = \int_{-\infty}^{\infty} |X(f)|^2 df$
2. $E_y = \frac{1}{2\pi}\int_{-\infty}^{\infty} |Y(\omega)|^2 d\omega = \int_{-\infty}^{\infty} |Y(f)|^2 df$

$$|Y(\omega)|^2 = |H(\omega)|^2 |X(\omega)|^2$$

$$|Y(f)|^2 = |H(f)|^2 |X(f)|^2$$

Eigenfunction and eigenvalue of LTI system:

Defined only for LTI systems

Real to complex or I

$$H(\omega) = \int_{-\infty}^{\infty} h(t)e^{-j\omega t}dt$$

$$\boxed{H(-\omega) = \int_{-\infty}^{\infty} h(t)e^{j\omega t}dt}$$

$$\boxed{x(t)} \longrightarrow \overset{LTI}{\boxed{h(t)}} \longrightarrow Y(t) = k.x(t)$$

1. x(t) is called eigenfunction of LTI s/s
2. K: Eigenvalue of LTI s/s → real/ complex / imaginary

Case a) Let system is fixed:

Case i. $h(t) = K\delta(t)$

$$x(t) \longrightarrow \boxed{h(t) = K\delta(t)} \longrightarrow$$

$$y(t) = x(t) * h(t) \qquad y(t) = kx(t)$$

Note: When h(t) is an impulse at the origin, then every x(t) is an eigenfunction.

Case ii. $h(t) = k\delta(t - T)$

$$\xrightarrow{x(t)} \boxed{h(t)} \longrightarrow y(t) = Kx(t - T) = Kx(t)$$

is periodic with T.

Note: $h(t) = K\delta(t - T)$ then every x(t) with period T will become eigenfunction.

Case b) When system is not fixed, but input is fixed.

Laplace Tx:

$$H(s)\int_{-\infty}^{\infty} h(t)e^{-st}dt \qquad s = \sigma + j\omega$$

$$h(t) \xleftrightarrow{L.T.} H(s)$$

complex variable

1. $x(t) = e^{s_o t} \longrightarrow \boxed{H(s)} \longrightarrow y(t)$

   $= e^{s_o t} \underbrace{H(s_o)}_{\substack{scalor \\ or \\ Engen\ value}}$

   $e^{s_o t}$ – *Generalised eigen function*

$$y(t)=\int_{-\infty}^{\infty}x(t-\tau)h(\tau)d\tau$$

$$y(t)=\int_{-\infty}^{\infty}e^{s_o t}e^{-s_o\tau}h(\tau)d\tau$$

$$y(t)=e^{s_o t}\int_{-\infty}^{\infty}h(\tau)e^{-s_o\tau}d\tau=e^{s_o t}H(s_o)$$

$$e^{s_o t}\longrightarrow H(s)\longrightarrow e^{s_o t}H(s_o)$$

$$e^{\sigma_o t}\longrightarrow H(s)\longrightarrow e^{\sigma_o t}H(\sigma_o)$$

$$e^{j\omega_o t}\longrightarrow H(s)\longrightarrow e^{j\omega_o t}H(j\omega_o)$$

$$1e^{j0(t)}\longrightarrow H(s)\longrightarrow 1H(0)$$

2. $e^{-s_o t}\longrightarrow H(s)\longrightarrow e^{-s_o t}H(-s_o)$

$$e^{-\sigma_o t}\longrightarrow H(s)\longrightarrow e^{-\sigma_o t}H(-\sigma_o)$$

$$e^{-j\omega_o t}\longrightarrow H(s)\longrightarrow e^{-j\omega_o t}H(-j\omega_o)$$

$$1\longrightarrow H(s)\longrightarrow H(0)$$

3. $\underbrace{x(t)}_{\text{Eigen Function}}=e^{j\omega_o t}\longrightarrow H(\omega)$

$$\longrightarrow y(t)=e^{j\omega_o t}\underbrace{H(\omega_o)}_{\text{Eigenvalue}}$$

$$y(t)=\int_{-\infty}^{\infty}x(t-\tau)h(\tau)d\tau$$

$$y(t)=\int_{-\infty}^{\infty}e^{j\omega_o t}e^{-j\omega_o\tau}h(\tau)d\tau$$

$$y(t)=e^{j\omega_o t}\int_{-\infty}^{\infty}h(\tau)e^{-j\omega_o\tau}d\tau=e^{j\omega_o t}H(\omega_o)$$

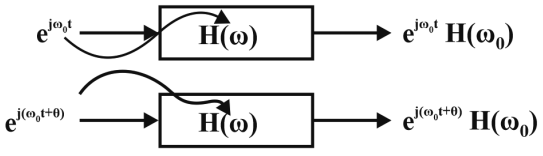


Fig. 5.56

1. $\longrightarrow H(\omega)\longrightarrow 1H(0)$

$$e^{-j\omega_o t}\longrightarrow H(\omega)\longrightarrow e^{-j\omega_o t}H(-\omega_o)$$

$$e^{-j(\omega_o t+\theta)}\longrightarrow H(\omega)\longrightarrow e^{-j\omega_o t}H(-\omega_o)e^{-j\theta}$$

$$\frac{e^{-j(\omega_o t+\theta)}}{e^{-j\omega_o t}e^{-j\theta}}\longrightarrow H(\omega)\longrightarrow e^{-j\omega_o t}H(-\omega_o)e^{-j\theta}$$

$$e^{j2\pi f_o t}\longrightarrow H(f)\longrightarrow e^{j2\pi f_o t}H(f_o)$$

$$e^{j(2\pi f_o t+\theta)}\longrightarrow H(f)\longrightarrow e^{j(2\pi f_o t+\theta)}H(f_0)$$

$$e^{-j2\pi f_o t}\longrightarrow H(f)\longrightarrow e^{-j(2\pi f_o t+\theta)}H(-f_o)$$

$$e^{-j(2\pi f_o t+\theta)}\longrightarrow H(f)\longrightarrow e^{-j(2\pi f_o t+\theta)}H(-f_o)$$

21. (*i*) $e^{-2t}u(t)\longrightarrow H(s)=\dfrac{1}{S+3}\longrightarrow y(t)$

$$=e^{-2t}u(t)*e^{-3t}u(t)$$

(*ii*) $e^{-2t}\xrightarrow[\text{Eigen function}]{}H(s)=\dfrac{1}{S+3}$

$$\longrightarrow y(t)=e^{-2t}H(-2)=e^{-2t}$$

Response of an LTI System for Sinusoidal Input

Case 1: h(t) is even: $\underset{EVEN}{h(t)}\longleftrightarrow\underset{EVEN}{H(\omega)}$

$$x(t)=A\cos\omega_o t\xrightarrow[H(\omega)]{LTI}\longrightarrow y(t)=?$$

$$\frac{Ae^{j\omega_o t}}{2}\longrightarrow H(\omega)\longrightarrow\frac{A}{2}e^{j\omega_o t}H(\omega_o)$$

$$\frac{Ae^{-j\omega_o t}}{2}\longrightarrow H(\omega)\longrightarrow\frac{A}{2}e^{-j\omega_o t}H(-\omega_o)$$

$$x(t)=A\cos(\omega_o t)\xrightarrow{EVEN}H(\omega)$$

$$\longrightarrow\frac{A}{2}e^{j\omega_o t}H(\omega_o)+\frac{A}{2}e^{-j\omega_o t}H(-\omega_o)$$

$$\{A\cos\omega_o t\}H(\omega_o)$$

Conclusions: h(t): Even

$$A\cos\omega_o t\longrightarrow H(\omega)\longrightarrow\{A\cos\omega_o t\}H(\omega_o)$$

$$A\cos(\omega_o t+\theta)\longrightarrow H(\omega)$$

$$\longrightarrow\{A\cos\omega_o t+\theta\}H(\omega_o)$$

$A\cos 2\pi f_o t \longrightarrow H(f)$

$\longrightarrow \{A\cos 2\pi f_o t\} H(f_o)$

$A\cos(2\pi f_o t + \theta) \longrightarrow H(f)$

$\longrightarrow \{A\cos(2\pi f_o t + \theta)\} H(f_o)$

h(t): even

$A\sin\omega_o t \longrightarrow H(\omega) \longrightarrow \{A\sin\omega_o t\} H(\omega_o)$

$A\sin(\omega_o t + \theta) \longrightarrow H(\omega)$

$\longrightarrow \{A\sin(\omega_o t + \theta)\} H(\omega_o)$

$A\sin 2\pi f_o t \longrightarrow H(f)$

$\longrightarrow \{A\sin 2\pi f_o t\} H(f_o)$

$A\sin(2\pi f_o t + \theta) \longrightarrow H(f)$

$\longrightarrow \{A\sin 2\pi f_o t + \theta\} H(f_o)$

Note: If h(t) is even:

1. $A\cos(\omega_0 t + \theta), A\sin(\omega_0 t + \theta)$ will be eigenfunction with same eigenvalue $H(\omega_0)$.
2. $H(\omega_0)$ is not necessarily real.
3. $A\cos(2\pi f_0 t + \theta), A\sin(2\pi f_0 t + \theta)$ will be eigenfunction with same eigenvalue $H(f_0)$.
4. $H(f_0)$ is not necessarily real.

Case 2: $h(t)$: Real $+ EVEN \longleftrightarrow$ $H(\omega)$: Real + EVEN

a) Both $A\cos(2\pi f_0 t + \theta), A\sin(2\pi f_0 t + \theta)$ will be eigenfunction with same real eigenvalue $H(f_0)$.

b) Both $A\cos(\omega_0 t + \theta), A\sin(\omega_0 t + \theta)$ will be eigenfunction with same real eigenvalue $H(\omega_0)$.

Case 3: $h(t)$ is real $\longleftrightarrow H(\omega)$: Conjugate symmetry

$|H(-\omega)| = |H(\omega)| : EVEN$

$\angle H(-\omega) = -\angle H(\omega) : ODD$

$x(t) = A\cos(\omega_0 t + \theta) \longrightarrow H(\omega) \longrightarrow y(t)$

$\frac{A}{2} e^{j(\omega_0 t + \theta)} \longrightarrow \boxed{H(\omega)} \longrightarrow \frac{A}{2} H(\omega_0) e^{j(\omega_0 t + \theta)}$

$\frac{A}{2} e^{-j(\omega_0 t + \theta)} \longrightarrow \boxed{H(\omega)} \longrightarrow \frac{A}{2} H(-\omega_0) e^{-j(\omega_0 t + \theta)}$

$x(t) = A\cos(\omega_0 t + \theta) \longrightarrow y(t)$

$= \frac{A}{2} H(\omega_0) e^{j(\omega_0 t + \theta)} + y(t)$

$= \frac{A}{2} H(-\omega_0) e^{-j(\omega_0 t + \theta)}$

$y(t) = \frac{A}{2} |H(\omega_0)| e^{j\angle H(\omega_0)} e^{j(\omega_0 t + \theta)}$

$+ \frac{A}{2} |H(-\omega_0)| e^{j\angle H(-\omega_0)} e^{-j\angle H(\omega_0 t + \theta)}$

$y(t) = \frac{A}{2} \underline{|H(\omega_0)|} e^{j[\omega_0 t + \theta - \angle H(\omega_0)]}$

$+ \frac{A}{2} \underline{|H(-\omega_0)|} e^{-j\left[\omega_0 t + \theta - \underbrace{\angle H(-\omega_0)}_{+\angle H(\omega_0)}\right]}$

$$\boxed{\begin{aligned} |H(-\omega)| &= |H(\omega)| \\ \angle H(-\omega) &= -\angle H(\omega) \end{aligned}}$$

$y(t) = A|H(\omega_0)|\cos(\omega_0 t + \angle H(\omega_0))$

Conclusions:

h(t): Real

1. $A\cos(\omega_0 t + \theta) \longrightarrow \underbrace{\boxed{H(\omega)}}_{H(\omega_0) = |H(\omega_0)| \angle H(\omega_0)}$

$\longrightarrow A|H(\omega_0)|\cos(\omega_0 t + \theta + \angle H(\omega_0))$

2. $A\cos(2\pi f_0 t + \theta) \longrightarrow \underbrace{\boxed{H(f)}}_{H(f_0) = |H(f_0)| \angle H(f_0)}$

$\longrightarrow A|H(f_0)|\cos(2\pi f_0 t + \theta + \angle H(f_0))$

3. $A\sin(\omega_0 t + \theta) \longrightarrow \boxed{H(\omega)}$

$\longrightarrow A|H(\omega_0)|\sin(\omega_0 t + \theta + \angle H(\omega_0))$

4. $A\sin(2\pi f_0 t + \theta) \longrightarrow \boxed{H(f)}$

$\longrightarrow A|H(f_0)|\sin(2\pi f_0 t + \theta + \angle H(f_0))$

Note: h(t): Real

Sinusoids will not be eigenfunction

22. $2\sin(3t+45°) \longrightarrow \boxed{h(t)=e^{-2t}u(t)}$

$\longrightarrow y(t)=?$

$h(t)$: Real $\longleftrightarrow H(\omega)$

Solution:

$\omega_0 = 3$

m.1 $y(t)=x(t)*h(t)$

m.2 $Y(\omega)=X(\omega)H(\omega)$

$y(t) \leftrightarrow Y(\omega)$

$y(t)=2\left|H(\omega_0)\right|\sin\left(3t+45°+\angle H(\omega_0)\right)$

$h(t)=e^{-2t}u(t) \longleftrightarrow H(\omega)=\dfrac{1}{2+j\omega}$

$H(\omega)=\dfrac{1}{2+j3} \rightarrow \begin{matrix} \rightarrow |H(\omega_0)| = \dfrac{1}{\sqrt{13}} \\ \rightarrow \angle H(\omega_0) = -\tan^{-3/2} \end{matrix}$

Questions Based on Filtering

23. $x(t)=4\cos(200\pi t)+8\cos(400\pi t)$.

$x(t) \longrightarrow h(t) \longrightarrow y(t)$

$$h(t)=\begin{cases} \dfrac{2\sin(300\pi t)}{\pi t} & : t \neq 0 \\ 600 & : t \neq 0 \end{cases}$$

a) Frequencies on y(t)

b) Calculate y(t) = ?

Solution:

Method 1:

$$x(t)=4\cos\left(\underbrace{200\pi t}_{200\pi}\right)+8\cos\left(\overbrace{400\pi t}^{400\pi}\right)$$

$$h(t)=\frac{2\sin(300\pi t)}{\pi t} \longleftrightarrow$$

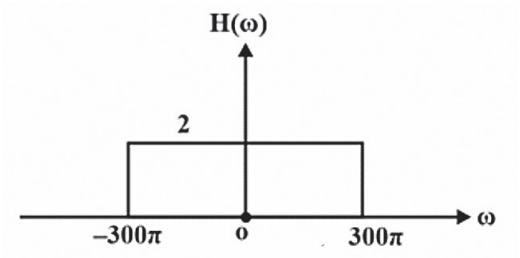


Fig. 5.57

$$\xrightarrow[\substack{4\cos(200\pi t)\\+\\8\cos(400\pi t).}]{x(t)} \boxed{h(t) \rightarrow H(\omega)}$$

$\longrightarrow y(t)=\{4\cos(200\pi t)\}H(\omega_0)$

$y(t)=8\cos(200\pi t)$

Method 2:

$x(t) \longrightarrow \boxed{h(t)} \longrightarrow y(t)$

S. 1 $Y(\omega)=X(\omega)H(\omega)$

S. 2 $y(t) \longleftrightarrow Y(\omega)$

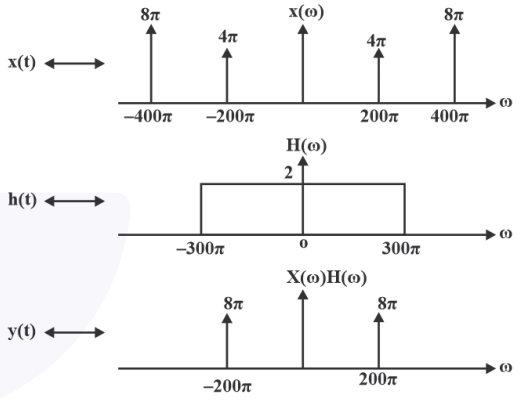


Fig. 5.58

$Y(\omega)=\pi\left[8\delta(\omega-200\pi t)+8\delta(\omega+200\pi t)\right]$

$\cos\omega_0 t \longleftrightarrow \pi\delta(\omega-\omega_0)+\pi\delta(\omega+\omega_0)$

$Y(\omega) \longleftrightarrow y(t)=8\cos(200\pi t)$

$1 \longleftrightarrow 2\pi\delta(\omega)$

$e^{j200\pi t} \longleftrightarrow 2\pi\delta(\omega-200\pi t)$

$e^{-j200\pi t} \longleftrightarrow 2\pi\delta(\omega+200\pi t)$

24. $x(t)=\cos 60\pi t+\sin 30\pi t$
$+\cos 30\pi t \times \cos 10\pi t$

$x(t) \longrightarrow \boxed{h(t)=\dfrac{2\sin 20\pi t}{\pi t}}$

$\longrightarrow y(t)=?$

Solution:

$$x(t) = \cos 60\pi t + \sin 30\pi t + \frac{1}{2}\cos(40\pi t) + \frac{1}{2}\cos(20\pi t)$$

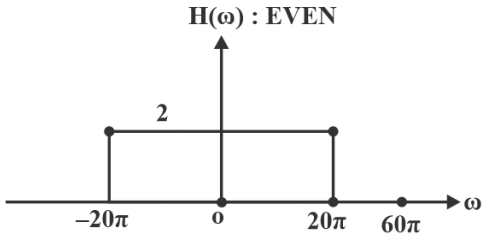


Fig. 5.59

$$x(t) \longrightarrow \boxed{H(\omega)} \longrightarrow y(t)$$

$$= \frac{1}{2}\cos(20\pi t)H(20\pi)$$

$$= \cos(20\pi t)$$

Systems with periodic input:

$$x(t) \longrightarrow \boxed{H(\omega)} \longrightarrow y(t)$$

Let x (t) is periodic signal

$$\boxed{\text{Relation between } C_k \text{ and } d_k = ?}$$

$$x(t) \xrightarrow{FSC} C_k$$

→ y (t) is also periodic

$$y(t) \xrightarrow{FSC} d_k$$

$$x(t) = \sum_{n=-\infty}^{\infty} C_n e^{jn\omega_0 t} = +\cdots\cdot C_{-1}e^{-j\omega_0 t} + C_0 + C_1 e^{j\omega_0 t} + \cdots\cdot$$

$$C_{-1}e^{-j\omega_0 t} \xrightarrow{LTI} \boxed{H(\omega)} \longrightarrow C_{-1}e^{-j\omega_0 t}H(-\omega_0)$$

$$C_0 \xrightarrow{LTI} \boxed{H(\omega)} \longrightarrow C_0 H(0)$$

$$C_1 e^{j\omega_0 t} \xrightarrow{LTI} \boxed{H(\omega)} \longrightarrow C_1 e^{j\omega_0 t}H(\omega_0)$$

$$\underbrace{C_n}\, e^{jn\omega_0 t} \longrightarrow \boxed{H(\omega)}$$

$$x(t) = \sum_{n=-\infty}^{\infty} \longrightarrow y(t) = \sum_{n=-\infty}^{\infty} \underbrace{C_n H(n\omega_0)}\, e^{jn\omega_0 t}$$

$$\boxed{\begin{aligned} x(t) &\longleftrightarrow C_n \\ y(t) &\longleftrightarrow dn = C_n H(n\omega_0) \end{aligned}}$$

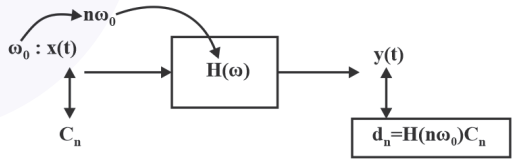


Fig. 5.60

Level 1

25. $x(t) = 3\cos(10\pi t) \longrightarrow \boxed{H(s) = \frac{s-1}{s+1}} \longrightarrow y(t)$

FSC of y (t) = ?

Solution:

$$\omega_0 = 10\pi: \quad x(t) = 3\cos(10\pi t)$$

$$\longleftrightarrow C_n = \begin{cases} \frac{3}{2} : n = 1 \\ \frac{3}{2} : n = -1 \end{cases}$$

$$y(t) \xleftrightarrow{FSC} d_n = C_n H(n\omega_0)$$

$$H(s) = \frac{s-1}{s+1}$$

$$H(\omega) = \frac{j\omega - 1}{j\omega + 1}$$

$$H(n\omega_0) = \frac{jn\omega_0 - 1}{jn\omega_0 + 1} = \left\{\frac{jn10\pi - 1}{jn10\pi + 1}\right\}$$

$$d_n = C_n \left\{\frac{jn10\pi - 1}{jn10\pi + 1}\right\}$$

$$\boxed{d_1 = C_1 H(n\omega_0) = \frac{3}{2}\left\{\frac{j10\pi - 1}{j10\pi + 1}\right\}}$$

$$\boxed{d_{-1} = C_{-1}H(-n\omega_0) = \frac{3}{2}\left\{\frac{1 + j10\pi}{1 - j10\pi}\right\}}$$

$$y(t)\, Power : \sum_{n=-\infty}^{\infty} |d_n|^2 = |d_1|^2 + |d_{-1}|^2 = \frac{9}{2} watt$$

Bandwidth:

Bandwidth of signal

1. Absoluted BW:

$x(t): BW = ?$

$x(t) \longleftrightarrow X(\omega) \longrightarrow |X(\omega)|$ vs ω

BW = [Highest +ve frequency at which component is present]

[Lowest +ve frequency at which component is present]

2. 3 – dB BW or 50% energy BW

$x(t) \longrightarrow BW = ?$

$x(t) \longrightarrow X(\omega): \longrightarrow |X(\omega)|$

$|X(\omega_c)| = \dfrac{|X(\omega)|_{max}}{\sqrt{2}} \longrightarrow \omega_c$

$3dB \quad BW = (\omega_c - 0) = \omega_c$: rad / sec

Solutions LEVEL - 1

22. $x(t) = e^{-2t}u(t)$: $3dB\ BW = ?$

Solution:

Method 1: $X(\omega) = \dfrac{1}{2 + j\omega}$

$|X(\omega)| = \dfrac{1}{\sqrt{4+\omega^2}}$: $|X(\omega)|_{max} = \dfrac{1}{2}$: $\omega = 0$

$|X(\omega_c)| = \dfrac{|X(\omega)|_{max}}{\sqrt{2}}$

$\dfrac{1}{\sqrt{4+\omega_c^2}} = \dfrac{1}{2\sqrt{2}} \longrightarrow 4 + \omega_c^2$

$= 8 \quad \omega_c = 2rad$ / sec

$3dB \quad BW = \omega_c - 0 = 2\,rad$ / sec

Method 2: $x(t) = e^{-2t}u(t) \longleftrightarrow X(\omega) = \dfrac{1}{2 + j\omega}$

$E_x = \dfrac{1}{2a} = \dfrac{1}{4}$: $Total\ energy$

$\dfrac{1}{2\pi}\int_{-\omega_c}^{\omega_c} |X(\omega)|^2\, d\omega = 50\%\left\{\dfrac{1}{4}\right\}$

$\dfrac{1}{2\pi}\int_{-\omega_c}^{\omega_c} \dfrac{1}{4+\omega^2} d\omega = \dfrac{1}{8}$

$\dfrac{1}{2}\left\{\tan^{-1}\dfrac{\omega}{2}\right\}_{-\omega_c}^{\omega_c} = \dfrac{2\pi}{8}$

$\dfrac{1}{2} \times 2\tan^{-1}\dfrac{\omega_c}{2} = \dfrac{\pi}{4}$

$\dfrac{\omega_c}{2} = 1$

$\omega_c = 2rad$ / sec

BW of system: BW of signal h (t)

1. Let the transfer function of a network be $H(f) = |H(f)|e^{j\theta(f)} = 2e^{-j4\pi f}$. If a signal $x(t)$ is applied to such a network, the output $Y(t)$ is given by:

(A) $2x(t)$

(B) $x(t-2)$

(C) $2x(t-2)$

(D) $2x(t-4\pi)$

2. Which one of the following is correct for Parseval equality?

(A) $\int_{-\infty}^{+\infty} |x(t)|^2\, dt = \dfrac{1}{2\pi}\int_{-\infty}^{+\infty} |X(j\omega)|^2\, d\omega$

(B) $\int_{-\infty}^{+\infty} |x(t)|^2\, dt = \dfrac{1}{2\pi}\int_{0}^{+\infty} |X(j\omega)|^2\, d\omega$

(C) $\int_{-\infty}^{-\infty} |x(t)|^2\, dt = \dfrac{1}{\pi}\int_{0}^{+\infty} |X(j\omega)|^{1/2}\, d\omega$

(D) $\int_{-\infty}^{-\infty} |x(t)|^2\, dt = \dfrac{4}{3\pi}\int_{-\infty}^{+\infty} |X(j\omega)|^2\, d\omega$

1. The Fourier transform of a rectangular pulse of a period

$t = -\frac{T}{2}$ to $t = \frac{T}{2}$ is

(A) A sinc function

(B) A sine function

(C) A cosine function

(D) A sine–squared function

2. The Fourier transform of a double–sided exponential signal $x(t) = e^{-b|t|}$

(A) Is $2b / (b^2 + \omega^2)$

(B) Is $\frac{e^{-j\tan^{-1}(\omega/b)}}{(b^2 + \omega^2)}$

(C) Does not exist

(D) Exists only when it is single-sided

3. $X(\omega)$ is the Fourier transform of $x(t)$ shown below. The value of $\int_{-\infty}^{\infty} |X(\omega)|^2 \, d\omega$ (rounded off to two decimal places) is _____.

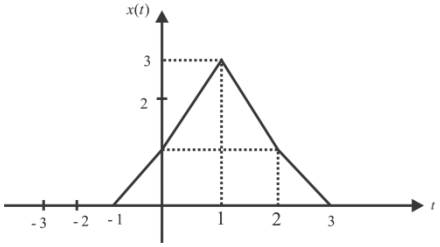


Fig. 5.61

Level 1					
1.	**C**	**2.**	**A**		
Level 2					
1.	**A**	**2.**	**A**	**3.**	**58.64**

1. (C)

Given, $H(f) = 2e^{-j4\pi f}$

And $x(t) \longleftrightarrow x(f)$

$\because$ output, $Y(t) = x(t) * h(t)$

Taking Fourier to transform, we get

$Y(f) = X(f)H(f)$

(according to the convolution property of F.T.)

$Y(f) = X(f)2e^{-j\pi\times 4f}$

$Y(f) = 2X(f)e^{-j2\pi f\times 2}$

Taking inverse F. T., we get

$Y(t) = 2F.T.^{-1}\left[X(f)e^{-j2\pi f\times 2}\right]$ ... (i)

We know that

Time-shifting property

$x\left(t - t_0\right) \longleftrightarrow x(f)e^{-j2\pi ft_0}$ ...(ii)

So from (1) and (2), we have

$Y(t) = 2x(t - 2)$

2. (A)

1. (A)

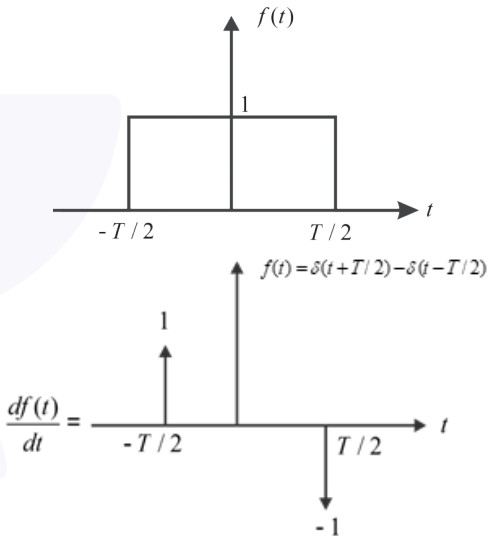


Fig. 5.62

$\because$ $f(t) \leftrightarrow F(\omega)$

By differentiation property in the time domain, we know.

$\frac{df(t)}{dt} \longleftrightarrow j\omega F(\omega)$

$\leftrightarrow 1/\, j\omega\left[e^{j\omega T/2} - e^{-j\omega T/2}\right]$

$\leftrightarrow \frac{1}{j\omega}\left[\begin{matrix}\cos\omega\frac{T}{2} + j\sin\omega\frac{T}{2} \\ -\cos\omega\frac{T}{2} + j\sin\omega\frac{T}{2}\end{matrix}\right]$

$f(t) \longleftrightarrow \frac{1}{j\omega}\left[2j\sin\frac{\omega T}{2}\right]$

$$= \frac{2}{\omega}\left[\sin\frac{\omega T}{2}\right] = 2 \times \frac{\sin\frac{\omega T}{2}}{\frac{\omega T}{2}} \times \frac{T}{2}$$

$$= \frac{2T}{2} Sa\left(\frac{\omega T}{2}\right)$$

$$f(t) \longleftrightarrow T\ Sa\left(\frac{\omega T}{2}\right)$$

2. (A)

$$x(t) = e^{-b|t|}$$

$$X(\omega) = \int_{-\infty}^{\infty} x(t)e^{-j\omega t}dt$$

$$= \int_{-\infty}^{\infty} e^{-b|t|}e^{-j\omega t}dt$$

$$= \int_{-\infty}^{0} e^{bt}e^{-j\omega t}dt + \int_{0}^{\infty} e^{-bt}e^{-j\omega t}dt$$

$$= \frac{1}{(b - j\omega)} e^{(b-j\omega)t}\Big|_{-\infty}^{0} + \frac{1}{-(b + j\omega)} e^{-(b+j\omega)t}\Big|_{0}^{-\infty}$$

$$= \frac{1}{b - j\omega} + \frac{1}{b + j\omega} = \frac{b + j\omega + b - j\omega}{b^2 + \omega^2}$$

$$\Rightarrow X(\omega) = \frac{2b}{b^2 + \omega^2}$$

3. (58.64)

Given: Signal $x(t)$ is shown in the figure.

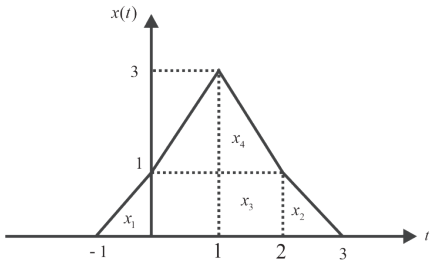


Fig. 5.63

Method 1

Using the Rayleigh energy theorem,

Energy of $x(t)$ is given by,

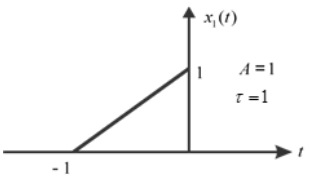
$$E_x = \int_{-\infty}^{\infty} |x(t)|^2 dt = \frac{1}{2\pi}\int_{-\infty}^{\infty} |X(\omega)|^2 d\omega$$

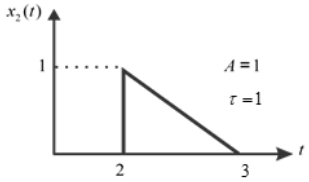
$$\therefore \quad I = \int_{-\infty}^{\infty} |X(\omega)|^2 d\omega = 2\pi E_x \qquad \text{... (i)}$$

To find the energy of $x(t)$, we can consider $x(t)$ to be combination of signals, $x_1(t), x_2(t), x_3(t)$ and $x_4(t)$ as shown in the figure. x_1 and x_2 are non-overlapping with x_3 and x_4 but x_3 and x_4 are overlapping with each other.

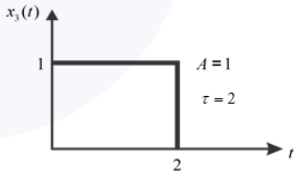
$$E_x = E\left[x_1(t)\right] + E\left[x_2(t)\right]$$

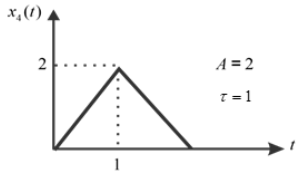
$$+E\left[\text{Combination of } x_3(t) \text{ and } x_4(t)\right] \quad \text{...(ii)}$$





$$E_{x_1} = \frac{1}{2} \times \frac{2A^2\tau}{3} = \frac{1}{3} \qquad E_{x_2} = \frac{1}{2} \times \frac{2A^2\tau}{3} = \frac{1}{3}$$

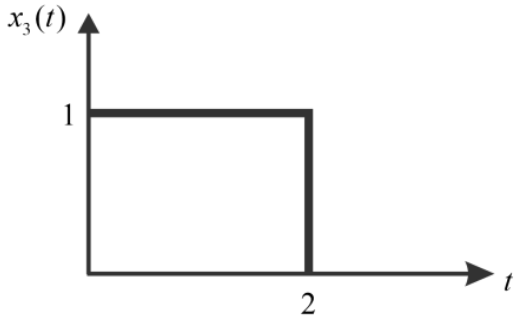


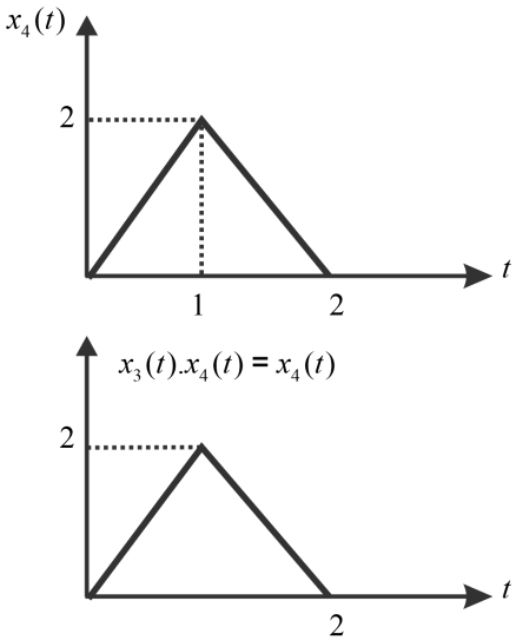


$$E_{x_3} = A^2\tau \qquad E_{x_4} = \frac{2A^2\tau}{3}$$

$$E_{x_3} = 1 \times 2 = 2 \qquad E_{x_4} = \frac{2 \times 4 \times 1}{3} = \frac{8}{3}$$

Energy in the combination of $x_3(t)$ and $x_4(t) = E_{x_3} + E_{x_4} + 2$ [Area under function $x_3(t)$. $x_4(t)$]





From figure,

$x_3(t).x_4(t) = x_4(t)$

Area $\left[x_3(t)\cdot x_4(t)\right] = \text{Area}\left[x_4(t)\right] = \frac{1}{2}\times 2\times 2 = 2$

Energy in the combination of $x_3(t)$ and $x_4(t)$

$= 2 + \frac{8}{3} + (2\times 2) = 6 + \frac{8}{3}$

From equation (ii), energy of $x(t)$ is

$E_x = \frac{1}{3} + \frac{1}{3} + 6 + \frac{8}{3} = \frac{1+1+18+8}{3} = \frac{28}{3}$

From equation (i), value of the given integral.

$I = \int_{-\infty}^{\infty} |X(\omega)|^2 \, d\omega = 2\pi \times \frac{28}{3} = 58.64$

Hence, the value of the integral is 58.64.